

Physics-informed versus data-driven surrogates for residual stress after wire drawing: accuracy, physical admissibility and behaviour outside the training range

Dmitriy Demin¹

HSE University,
11 Pokrovsky Boulevard,
Moscow, Russia
email: ddemin@hse.ru

Georgii Romanenko

HSE University,
11 Pokrovsky Boulevard,
Moscow, Russia
email: Georgii.v.romanenko@gmail.com

Residual stresses left by cold wire drawing govern the fatigue life and dimensional stability of the product, yet a finite-element evaluation is too slow for design-space exploration. We compare data-driven and physics-informed surrogates for the residual stress field of axisymmetric AISI 1020 drawing on a common dataset of 1756 finite-element simulations, cleaned from 2443 raw runs by removing friction-label duplicates, an adhered-to-die reduction level, and runs that did not complete the draw. Three one-dimensional families—a data-driven multilayer perceptron (MLP), a strong-form physics-informed neural network (PINN) and a weak-form variational PINN (VPINN)—share one architecture, one optimiser and the same training conditions and differ only in the loss, so that any difference is attributable to the physics term. Across thirteen interpolation and extrapolation splits the PINN has a lower error than the MLP on every split, but the margin is small (0.3–1.9 MPa at an 8–16 MPa error level) and does not survive a multiple-comparison correction; a data-scarcity advantage is not observed. Two backend implementations (PyTorch and JAX) are statistically indistinguishable in accuracy, so the cheaper one—here PyTorch, 1.65 times faster—should be preferred. The physics term earns its place elsewhere: it roughly halves the equilibrium and free-surface residuals of the predicted field, and it makes the surrogate markedly more robust to corrupted training labels (the PINN–MLP gap grows by 14.2 MPa per unit corrupted fraction). The finite-element model itself was not experimentally validated; the study therefore characterises the surrogates relative to their data source.

Keywords: physics-informed neural networks, wire drawing, residual stresses, variational PINN, finite element simulation, surrogate modelling

1 Introduction

Metal forming is a fundamental process in modern manufacturing, significantly impacting the final geometry, microstructure, and mechanical properties of manufactured products. Among various forming operations, wire drawing stands out as a widely employed technique, yet one that demands exceptional attention to surface quality, dimensional precision, and performance stability. The critical factor determining the dimensional stability, fatigue resistance, and durability of cold-drawn wire lies in the residual stresses generated during intense and non-uniform plastic deformation processes [1–12]. In particular, as demonstrated in [3] through precise finite-element analysis, the alternating loading patterns during multi-pass wire drawing result in intricate residual stress distributions that significantly influence the subsequent behavior of the wire. Studies [4,5] indicate that the die geometry and the conditions of lubrication during the processing of ZnAl15% alloy significantly affect the magnitude and distribution of residual stresses. For austenitic stainless steel X5CrNi18-10, as demonstrated in [10], it has been both experimentally and numerically verified that adjusting the inlet die angle and drawing speed allows for precise control over the residual stress state. However,

this approach necessitates extensive computational resources. In the production of superconducting wires based on Nb₃Sn, transverse deformations before heat treatment can lead to irreversible deterioration of critical current [1]. For high-strength, unalloyed steels like 30MnSi6, ensuring the stability of mechanical properties after bi-directional drawing is crucial [2]. This imposes strict limits on the acceptable level of residual stresses, making it a critical consideration in manufacturing processes. Thus, the capacity for swift and physically accurate prediction of residual stresses has become an indispensable aspect of contemporary drawing process design. Finite element analysis (FEM), rightfully regarded as the gold standard for stress-strain modelling during drawing, allows for highly accurate simulation of stress fields with the aid of axisymmetric and three-dimensional formulations. These advanced methods enable the intricate evolution of stress patterns to be meticulously reproduced, taking into account intricate contact interactions, substantial plastic deformations, material hardening, and thermal effects. However, the computational expense associated with a single FEM analysis remains prohibitively high, rendering it unsuitable for multi-parameter studies, rapid optimization, and real-time decision-making. This has spurred a surge in interest in surrogate models that promise orders-of-magnitude reduction in prediction time while maintaining acceptable levels of accuracy.

The advancements in machine learning (ML) technology have

¹Corresponding Author.

Version 1.18, September 21, 2026

opened up a whole new realm of possibilities for streamlining finite element analysis (FEA) in industrial settings [13–15]. By training artificial neural networks (ANNs) on representative datasets generated through FEA, these networks can almost instantaneously approximate highly nonlinear mappings between process parameters and stress fields. In the realm of wire drawing, this approach has been systematically explored in several studies [16–18]. One integrated scheme involving both FEA and ANNs [16] showcased the potential to predict residual stresses with remarkable accuracy. Moreover, the research conducted in [18] demonstrated that ANNs are capable of not only predicting residual stresses but also accurately reproducing accumulated strains in wires. Further in-depth investigations have revealed that when hyperparameters are meticulously optimised using tools such as Optuna [19], BayesianOpt, or Scikit-optimize, a multi-layer perceptron (MLP) demonstrates an R^2 coefficient of determination exceeding 0.99 for all components of the residual stress tensor in low-carbon steel wire. Similar patterns are observed in related domains. Convolutional neural networks prove effective in indentation analysis, simultaneously predicting non-equiaxial residual stresses and plastic flow properties [20,21]. For real-time estimation of three-dimensional residual stress distributions during machining, architectures based on the Incep-Net approach [22] are employed. Hybrid FEM-ANN-XGBoost models are utilised for predicting drawing forces [23]. A comprehensive review [15] underscores the prevailing trend towards integrating machine learning techniques for optimising manufacturing processes.

Nonetheless, purely data-driven models have a fundamental flaw: they cannot guarantee adherence to the laws of physics, specifically the equations of equilibrium, boundary conditions, and the governing principles of the mechanics of deformable materials. This issue becomes particularly critical when extending predictions beyond the scope of the training data, raising legitimate concerns about the physical validity of the predicted stress patterns. To address this limitation, the concept of Physics-Informed Neural Networks (PINNs) was introduced. The idea was first proposed in [24] and has undergone rapid development in recent years [25]. The core concept behind PINNs lies in integrating differential equations that describe the physical principles of a process directly into the loss function. This allows the neural network to be penalized not only for discrepancies with the data but also for violating conservation laws, boundary conditions, and initial conditions. As demonstrated in [26], this methodology ensures enhanced generalization capabilities and physical consistency in predictions for solid mechanics challenges. A comprehensive analysis [27] highlights the rapid expansion of physically informed machine learning in a wide range of manufacturing processes.

There are already successful examples of PINN applications in predicting residual stresses during ultrasonic surface treatment of 20CrNiMo steel [28]. Additionally, PINNs have been used to identify subsurface residual stresses in sliding friction processes [29,30] and assess durability under high-temperature creep conditions [31]. A comparative analysis of pure neural networks and physically regularized models applied to sheet metal stamping [32] demonstrated that PINN-like architectures significantly improve generalization even with limited training data. The convergence of hybrid finite-element-machine learning models and physics-driven machine learning approaches for predicting the properties of additive manufactured steels underscores a broader trend towards the integration of prior physical understanding into learning algorithms [33,34].

Despite the general progress made, the full potential of physical informed neural networks (PINNs) for predicting residual stresses after wire drawing remains largely unexplored. In a study [17] that demonstrated unprecedented accuracy with a purely data-driven approach, no explicit equilibrium constraints were imposed on the neural network model. Consequently, it is still not clear whether physical regularization improves prediction quality, or whether the rigidity imposed by equilibrium equations reduces accuracy on real-world datasets. Moreover, the existing literature lacks a sys-

tematic comparison of different PINN implementations, especially between the strong (differential) and weak (variational) formulations of physical laws for this class of problems. The present study seeks to bridge this gap by conducting a direct comparative analysis of three surrogate families on a unified dataset obtained through finite-element simulations of an axisymmetric wire-drawing process: a data-driven multi-layer perceptron (MLP), a strong-form physics-informed neural network (PINN), and a variational PINN (VPINN) built on the weak (integral) formulation of the equilibrium equations. The three families share one architecture, one optimiser and the same training budget and differ only in the composition of the loss, so that any difference in behaviour is attributable to the physics term rather than to the model or the training procedure.

Each family is studied in two tracks. In the one-dimensional track the surrogate maps the process parameters and the radial coordinate to the residual stress tensor along a representative section; in the extended track the axial coordinate is added as an input and the full two-dimensional field is predicted, so that both equilibrium equations can be evaluated on the prediction. The influence of the software back-end is examined separately, by training the same strong-form PINN in two independent implementations, PyTorch [35] and JAX/Flax [36]; a DeepXDE implementation [37] runs on the PyTorch back-end and differentiates with the same automatic-differentiation engine, so the comparison involves two differentiation engines rather than three. It is of paramount importance that all surrogates employ an identical representation of input features — the process parameters and the point coordinates — as well as a uniform output vector of the residual stress tensor components. A single group-aware data-partitioning scheme, in which all points of one finite-element computation are kept together on the same side of every split [38], and a shared set of quality indicators are used throughout.

This experimental design guarantees that any statistically significant differences in prediction accuracy can be attributed to the way physical knowledge is integrated and to the software implementation, excluding the influence of extraneous factors. Accuracy is assessed per component of the stress tensor and as a field-averaged macro-RMSE; differences between families are tested with a paired t -test over the random seeds within each split ($n = 5$ pairs), a Friedman test being used only as an auxiliary check across splits. The comparison against the finite-element source is made on a common held-out set of 261 finite-element computations, and the random interpolation split holds out 263 computations. The findings yield quantitatively supported recommendations for selecting a surrogate architecture for the prediction of residual stresses following drawing operations, and delineate the conditions under which strong and weak forms of physical regularisation are worthwhile in mechanical-engineering processes.

2 Materials and methods

2.1 Materials and process parameters. The process modelled is axisymmetric cold drawing of a wire through a conical die. The computational domain and the notation used for its boundaries are shown in Fig. 1. Five characteristic surfaces are distinguished: S_1 - is the axis of symmetry of the billet and of the finished product; S_2 - is the outlet surface, on which the drawing velocity is prescribed; S_3 - is the free lateral surface; S_4 - is the contact surface between the wire and the die; and S_5 - is the entry surface of the billet.

The half-angle of the working cone of the die is denoted α ; D_0 and D are the diameters of the billet and of the drawn wire; L is the length of the calibration land. The radial coordinate r is measured from the axis and the axial coordinate z along the direction of metal flow. The three geometric parameters of Table 1 are expressed through this notation: the reduction $Q = (r_0 - r_1)/r_0$ follows from D_0 and D , the deformation-zone half-angle is α , and the calibration ratio k relates the land length L to the outlet radius.

A series of axisymmetric FEM simulations of the wire-drawing process was carried out in Abaqus CAE [39]. The workpiece

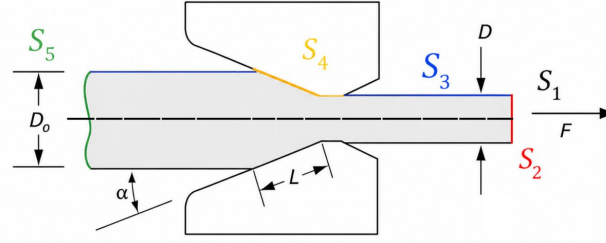


Fig. 1 Axisymmetric wire-drawing process. S_1 – axis of symmetry; S_2 – outlet face of the finished product; S_3 – free lateral surface; S_4 – contact surface with the die; S_5 – entry face of the billet; α – half-angle of the working cone of the die; D_0 and D – diameters of the billet and of the drawn wire; L – length of the calibration land

was a rectangular longitudinal section discretised with a CAX4RT element mesh. The geometry of the die was described by three parameters: the reduction of diameter (Q), the half-angle of the deformation zone (α), and the ratio between the calibration-section length and the outlet radius (k). Process parameters varied during the campaign were the Coulomb friction coefficient (μ) and the drawing velocity (v). Their ranges are summarised in Table 1.

Material behaviour is described by the Johnson–Cook model, with the constants of the input deck: $A = 213$ MPa, $B = 53$ MPa, $n = 0.345$, $m = 0.81$, melting temperature 1386°C and transition temperature 20°C for the thermal term, and $C = 0.055$ with a reference strain rate of 0.004 s^{-1} for the rate term. Elastic constants are $E = 190$ GPa and $\nu = 0.29$; the density is 7870 kg m^{-3} , the specific heat $470\text{ J kg}^{-1}\text{K}^{-1}$ and the inelastic heat fraction 0.9. The analysis is a coupled temperature–displacement one and is carried out in degrees Celsius, with an initial temperature of 20°C throughout.

All three arguments of the constitutive equation vary strongly across the section, so no single yield stress describes the pass. Over the four runs for which the strain, strain-rate and temperature histories were exported, the equivalent plastic strain runs from 0.034 on the axis to 0.46–0.58 at the surface, the strain rate from 3 to 99 s^{-1} , and the temperature from the initial 20°C to 66 – 95°C at the surface. Because the thermal exponent is below unity, the homologous temperature enters with disproportionate weight: a rise of 75 K gives a homologous temperature of 0.055 but a softening of 9.5%.

Where a single representative flow stress is needed – in the comparison with the analytical solutions of Section 3.1, which are energy methods – it is formed by averaging the constitutive equation along the deformation path of each material point, weighted by plastic work, and then over the section weighted by radius: $\sigma_f = \int \sigma(\epsilon, \dot{\epsilon}, T) d\epsilon_p / \int d\epsilon_p$. Evaluated on the exported histories this gives $\sigma_f = 300$ – 305 MPa, with a node-to-node spread of 260–324 MPa across the section. The same constitutive description was used in all simulations, so the resulting dataset reflects pure variation of geometry and process conditions. Spatial coordinates of mesh nodes and the four cylindrical components of the residual stress tensor were exported from the Abaqus Output Database (.odb) files through scripts written against the Abaqus CAE Python API. Only the nodes lying within 25–75% of the wire cross-section height were retained for surrogate-model training, because this region corresponds to the steady-state deformation regime and is most representative of the residual stress field of interest. Each retained set was downsampled to 20 radial points so that all simulations share the same spatial grid.

After removing friction-label duplicates, the highest reduction level $Q = 0.25$ (at which several runs ceased to be drawing as the billet adhered to the die) and runs that did not complete the draw, the cleaning detailed in Section 2.2 leaves 1756 valid simulations

Table 1 Ranges of process parameters in the FEM campaign.

Parameter	Values
Reduction Q	0.015, 0.025, 0.05, 0.10, 0.15, 0.20, 0.25
Calibration ratio k	0.0, 0.1, 0.3, 0.5, 0.75, 1.0
Die half-angle α , $^\circ$	4, 8, 12, 16, 20
Friction coefficient μ	0.025, 0.05, 0.10
Drawing velocity v , m/min	5, 10, 20, 40, 250

for surrogate-model training.

The modelling approach follows established practice in the numerical analysis of wire drawing: an axisymmetric finite-element discretisation, a Coulomb contact formulation at the die interface, and an elastic–plastic material description. Independent studies built on the same ingredients [3,6,7,10] report good agreement with measurements, which supports the approach; the present dataset itself was not compared with measurements on drawn wire, and the residual stress values are therefore interpreted with respect to the model that produced them. The internal consistency of that model, and its agreement with the analytical solutions for the drawing stress, are examined in Section 3.1.

2.2 Preparing Data for Machine Learning.

Exported fields and their limits.. Each report holds one frame of nodal values, the last one of the run: coordinates (r, z), logarithmic strains LE11, LE22, LE33, LE12, the von Mises and principal stresses, and the components S11, S22, S33, S12, which in the CAX convention are σ_{rr} , σ_{zz} , $\sigma_{\theta\theta}$, τ_{rz} . The targets are read directly from these columns (present in 2443 of 2443 files), and the identity of every stored field is verified against all quantities of the report, not only the four expected ones. The dataset was regenerated from the raw Abaqus reports; the four components (σ_{rr} , $\sigma_{\theta\theta}$, σ_{zz} , τ_{rz}) are read from the columns S.S11, S.S33, S.S22 and S.S12 respectively. Nodal averaging leaves a residual at the free surface: on the final dataset the mean $|\sigma_{rr}|$ at $r = R$ is 3.52 MPa and $|\tau_{rz}|$ 0.24 MPa. This is the accuracy with which the source itself satisfies the traction-free condition and is the baseline for every boundary-condition metric in Section 3.2.3. In the retained runs the exported cross-section is unloaded (median $|\int \sigma_{zz} dA|/A$ of 5.3 MPa against an rms σ_{zz} of 167 MPa), so the drawing force is not read from these cases.

Axial window.. Only nodes whose axial coordinate lies within 25–75 % of the axial extent of the wire are used (median captured fraction 0.4985); the window is the steady-state part of the drawn wire, of median length 57.5 mm at a radius of 16.2 mm

($L/R \approx 3.5$). The adequacy of this choice was examined component by component: for each component the share of the in-window variance attributable to the axial direction was measured, together with the loss incurred when the field is collapsed along z (Table 2). For the normal components these two quantities amount to 10–12 % and 2–4 % respectively, whereas for τ_{rz} they reach 40 % and 79 %: that component changes sign along z and therefore cancels under averaging. The shear component is thus intrinsically unpredictable from a one-dimensional representation, a property of the data rather than of any model (Fig. 4: inside the window the mean $|\tau_{rz}|$ of the shown run is 1.06 MPa against 11.0 MPa outside).

Coordinates in the field plots are dimensionless throughout. In Fig. 4 the radius is normalised by the billet radius, written R in the axis label of that figure, so that the drawn part of the wire ends at $r/R = 0.9$, and the axial coordinate by the length of the model; in Figs. 5 and 6 the radius is normalised by the surface radius of the same cross-section, so that $r/R = 1$ is the free surface at every axial position, and z by the window length. The radial axis is stretched relative to the axial one: at $L/R \approx 3.5$ – 7.7 an equal-aspect panel degenerates into a strip.

Profiles and grids.. For the main track the window nodes are sorted by radius and split into 20 equal-count bins; the bin radius and the four components are the medians within each bin. Radii are stored per case, and r is normalised by the outermost bin radius of the same case, so that $r = 1$ is the free surface by construction (median ratio of the outermost bin to the wire radius 0.996). Numerical effects inherent in the finite-element treatment of large plastic deformation displace individual nodes in the vicinity of the axis across the axis of symmetry, so that the radial coordinate recovered for them turns out slightly negative ($|r|/R \leq 0.054$); in all $1/r$ terms the radius is therefore clipped at $0.05R$. For the extended track the same nodes are binned on an 8×20 grid in (z, r) by the median; r is normalised by the surface radius of the same axial slice, z by the window ($z = 0$ entry, $z = 1$ exit). Targets stay in MPa; since the components differ by two orders of magnitude (mean $|\sigma|$ about 79 MPa for the normal components against 0.67 MPa for τ_{rz}), per-component errors are always reported next to macro-averaged ones.

Filtering.. In the further preparation of the data a part of the computed results had to be discarded, and this was necessary for several distinct reasons. Cases are removed whole, before any split, in the order of Table 3. The level $Q = 0.25$ is excluded entirely: there the process ceased to be drawing in several runs (the billet adhered to the die and the step degenerated into tension), and 17.5 % of its runs did not complete against 4.3–6.0 % at the other levels. Cases whose profile is bit-identical to another case differing only in the friction label are dropped, keeping one per group (323 of the 2443 raw cases, 13.2 %; 319 of them survive the two preceding filters and are counted in Table 3). Finally, runs that stopped before the wire had passed through the die are removed: their tail retains the billet radius R_0 and the die reference point lies inside the axial extent of the wire (153 of the 159 such runs on the raw grid). Where the undrawn part enters the window the surface radius varies along it (spread above 0.1 mm against a median of 0.0026 mm), and what the profile labels as free-surface σ_{rr} is contact pressure, down to -571.3 MPa against -19.7 MPa at worst in clean cases. These 92 cases (89 detected through the window plus 3 with the tail outside it) are removed before splitting: on the random split of the uncleaned set 73 of the 89 fell into the training pool, where the data term contradicts the traction-free condition.

The final dataset has 1756 cases: 35 120 radial points in the main track and 280 960 grid points in the extended track.

Group-aware splits.. For the training that follows, several scenarios of partitioning the initial sample were considered, each of them addressing a different question about the behaviour of the

surrogates. The runs carried out according to the design of Table 1 yield, once the filtering described above has been applied, configurations of points that are distinct within the window considered: the computations whose fields coincided were removed as duplicates. The partitioning is accordingly performed at the level of whole computations, so that the points belonging to one computation are never distributed between the training and the test set. A random 15 % hold-out measures interpolation only; to measure transfer beyond the training range, entire factor levels are held out (Table 4). Each extrapolation split is accompanied by two controls. The *matched* control withholds a random subset of the same size drawn from the whole pool, so that the difference $\Delta\text{RMSE} = \text{RMSE}(\text{extrap}) - \text{RMSE}(\text{matched})$ separates the cost of leaving the training range from the cost of a smaller training set; the *interior* control withholds an inner level of the same axis ($\alpha = 12^\circ$, $Q = 0.10$), which separates the cost of a gap inside the factor grid from the cost of leaving its range. A further partition, denoted `extrap_joint:corner` in the run log, is defined by a joint departure from the ranges of two factors, the die semi-angle α , i.e. the half-angle of the deformation zone, and the reduction Q : the region $\alpha = 20^\circ \wedge Q \geq 0.15$ is withheld as a whole. Along either of the two axes taken separately the test cases of this partition remain inside the training range, so that what is probed is a region of the joint factor space left uncovered by the design, and not extrapolation beyond the range of a single factor. A coverage report states, for each axis, whether the test range leaves the training range; transfer beyond the training range is claimed only for those axes for which it does. The partitions are fixed (seed 42); the five training seeds of Section 2.3 vary the initialisation, the batch order and the 15 % early-stopping subset carved from the training pool.

2.3 Preparing Machine Learning models. In preparing the machine-learning models whose comparative analysis is reported below, a single network architecture and a single optimiser were adopted for every family, and within each track all families were trained under the same conditions. The network and its inputs are described first, then the two tracks and the treatment of the axial coordinate, the loss function and its weak-form variant, the training protocol, the metrics and the physical audit, the two backends, the statistical treatment of the results and, last, the architecture and hyperparameters held fixed throughout.

Network and inputs.. All surrogates share one network and one optimiser, and the families compared with one another are trained under the same conditions: the same limit on the number of epochs, the same early-stopping criterion with the same patience, the same mini-batch size and batch-ordering rule, and the same seeds; the values, which differ between the main and the extended track, are stated in full in “Training protocol” below. The families differ only in the composition of the loss (Table 5, Figs. 2 and 3). In the main track the input is $\mathbf{x} = (Q, k, \alpha, \mu, v, r)$, with $r \in [0, 1]$ the radius normalised by the free-surface radius of the same case ($r = 1$ is the free surface). The radius is augmented by a Fourier embedding [40] $[\sin(2\pi kr), \cos(2\pi kr)]$, $k = 1, \dots, 4$; inputs and targets are standardised with training-set statistics; the body is a multilayer perceptron with three hidden layers of 96 tanh units and a linear head with four outputs $\hat{\sigma} = (\hat{\sigma}_{rr}, \hat{\sigma}_{\theta\theta}, \hat{\sigma}_{zz}, \hat{\tau}_{rz})$, de-standardised before entering any physical term. All arithmetic is in double precision in both backends. Each FEM case contributes 20 radial points, always kept in the same partition (group-aware splitting [38]).

Two tracks and the role of z .. The physics-informed families are built on the equations of equilibrium of an axisymmetric body, which are given below. The quantities entering them depend on both coordinates, r and z . In preparing the models, however, relations (1)–(2) undergo a number of changes, which are set out in

Table 2 Share of the variance along z inside the axial window and effect of collapsing the field along z by the median (mean absolute value, MPa).

Component	Share along z	Before collapse	After collapse
σ_{rr}	0.114	47.9	46.6
$\sigma_{\theta\theta}$	0.116	56.6	54.4
σ_{zz}	0.098	140.2	137.6
τ_{rz}	0.397	3.22	0.68

Table 3 Case-level filtering applied before splitting.

Step	Removed	Remaining
Raw reports		2443
Level $Q = 0.25$ excluded	275	2168
Friction label outside the design (file-name typo)	1	2167
Bit-identical duplicates differing only in μ label	319	1848
Runs that did not complete the pass through the die	92	1756

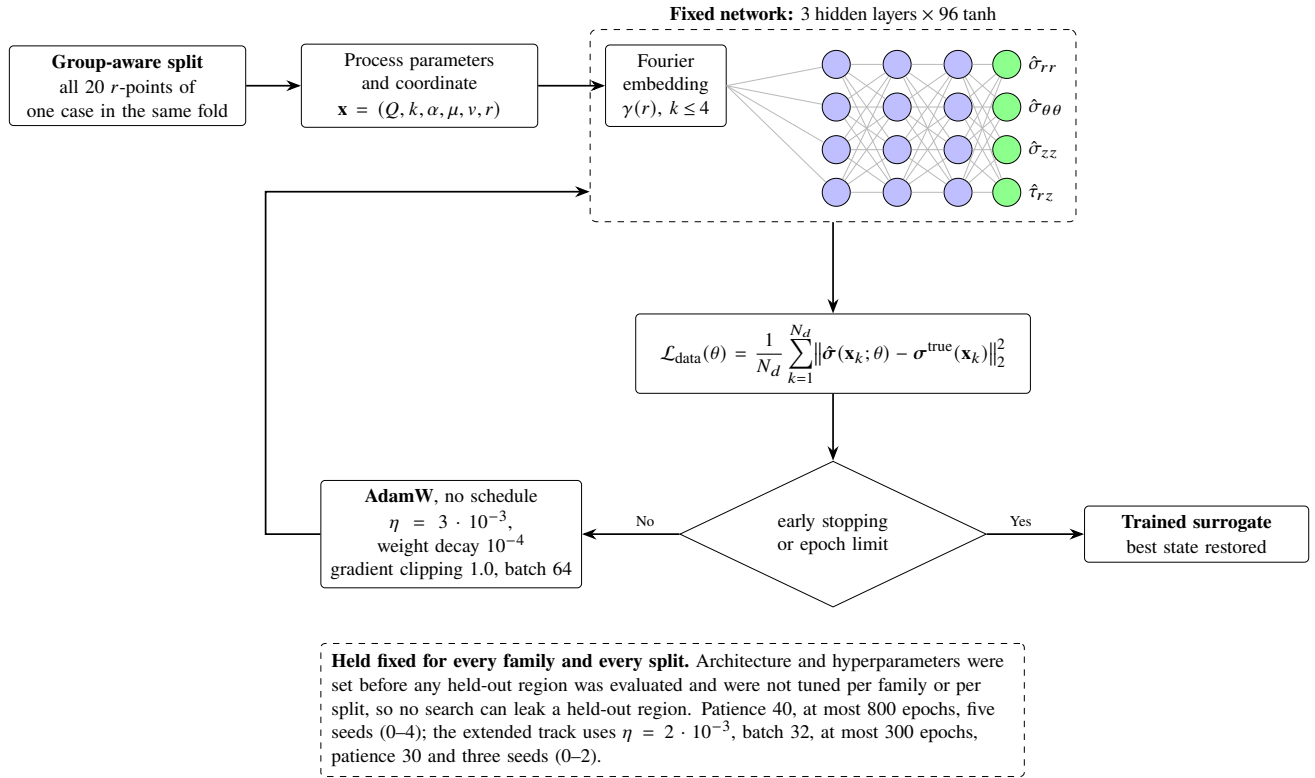


Table 4 The 13 case-level splits of the 1756 cases. n_{train} is the training pool, including the early-stopping subset.

Split	Held-out cases	Kind	n_{train}	n_{test}
random	random 15 %	interpolation	1493	263
extrap:alpha_max	$\alpha = 20^\circ$	extrapolation, α	1377	379
extrap:Q_high	$Q = 0.20$	extrapolation, Q	1484	272
extrap:k_max	$k = 1.0$	extrapolation, k	1466	290
extrap:v_max	$v = 250$ m/min	extrapolation, v	1255	501
extrap_joint:corner	$\alpha = 20^\circ \wedge Q \geq 0.15$	gap in the joint (α, Q) range	1642	114
interp:alpha_mid	$\alpha = 12^\circ$	interior control	1416	340
interp:Q_mid	$Q = 0.10$	interior control	1438	318
matched:alpha_max	random, same size	matched control	1377	379
matched:Q_high	random, same size	matched control	1484	272
matched:k_max	random, same size	matched control	1466	290
matched:v_max	random, same size	matched control	1255	501
matched:corner	random, same size	matched control	1642	114

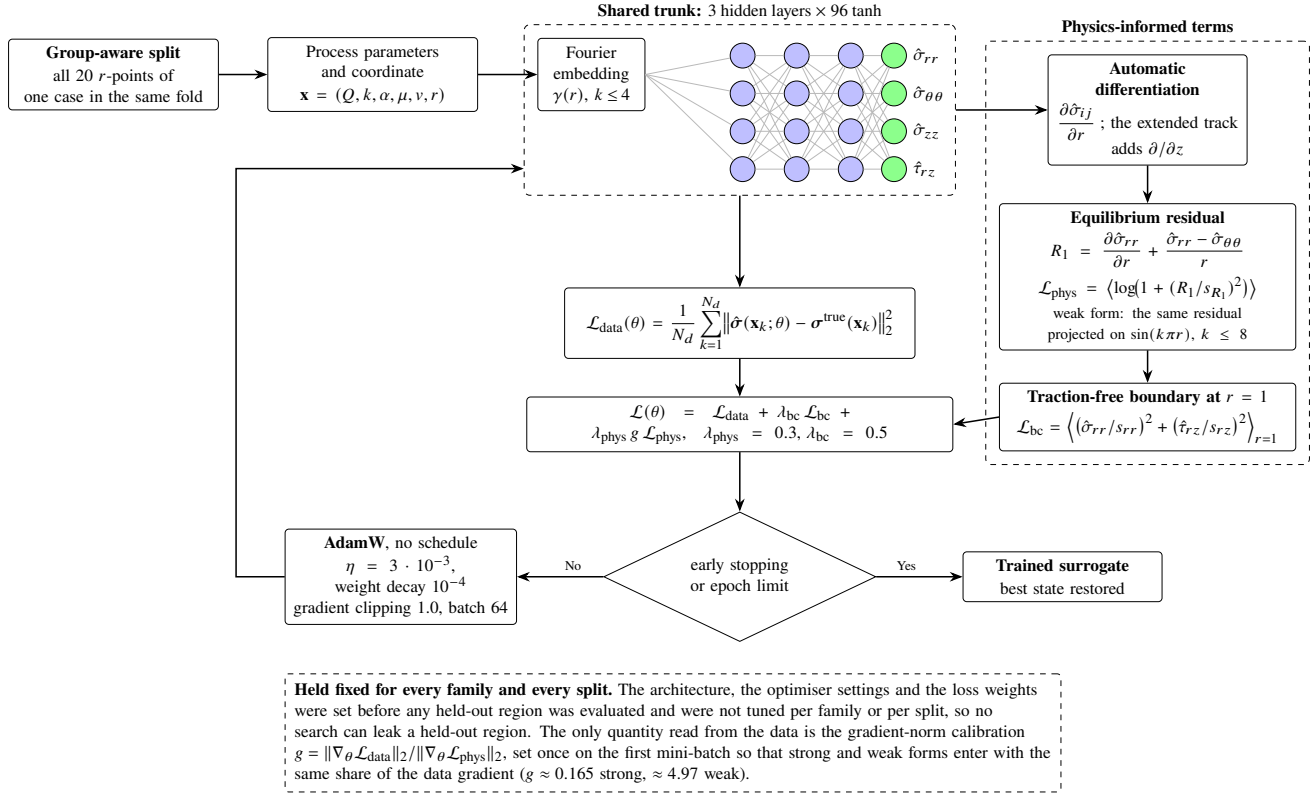


Fig. 3 Training of the physics-informed surrogate: shared trunk, equilibrium residual and traction-free boundary term in the loss. The loss weights are fixed a priori; the only quantity read from the data is the gradient-norm calibration g of Eq. (8), set once on the first mini-batch.

which the FEM data contradict (median relative range of $r \tau_{rz}$ along the radius 3.15). The term exists in the code under its true name (zero-shear penalty, λ_{shear}) and is switched off, $\lambda_{\text{shear}} = 0$, in every run reported here.

In the extended track the axial coordinate is added, $\mathbf{x} = (Q, k, \alpha, \mu, v, z, r)$; the field is sampled on an 8×20 (z, r) grid inside the same window, one median per cell, without collapsing the axial direction; z is normalised to the window ($0 = \text{inlet}$, $1 = \text{outlet}$) and r to the surface radius of the same axial section. Both equations are then written in full, and three families are compared: `mlp2d` (no physics), `pin2d_reduced` (R_r without $\partial \tau_{rz} / \partial z$, exactly the operator of the main track) and `pin2d_full` (both residuals). Physical derivatives follow from the chain rule, $\partial / \partial r_{\text{phys}} = R^{-1} \partial / \partial r$, $\partial / \partial z_{\text{phys}} = L^{-1} \partial / \partial z$, with R and L the surface radius and window length of the case; the $1/r$ factor uses r clipped at $0.05R$, since nodal radii near the axis carry noise (worst

$|r|/R = 0.054$).

Loss function. Every physics-informed family minimises

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{phys}} g \mathcal{L}_{\text{phys}}, \quad (5)$$

with $\mathcal{L}_{\text{data}}$ the mean squared error in standardised output space. The boundary term enforces the traction-free condition at the free surface,

$$\mathcal{L}_{\text{bc}} = \left\langle \left(\sigma_{rr}(r=1)/s_{rr} \right)^2 + \left(\tau_{rz}(r=1)/s_{rz} \right)^2 \right\rangle_{\text{batch}}, \quad (6)$$

where s_c is the training standard deviation of component c ; the condition is imposed and later audited at the same point, $r = 1$ (at every z in the extended track). The physics term is a $\log(1 + x^2)$

transform of the dimensionless residual,

$$\mathcal{L}_{\text{phys}} = \left\langle \log(1 + (R/s_R)^2) \right\rangle, \quad (7)$$

evaluated at the 20 data radii (collocation points coincide with data points), with scales $s_{R_1} = (s_{rr} + s_{\theta\theta})/R_{\text{char}}$ for the radial and $s_{R_z} = s_{zz}/L_{\text{char}} + s_{rz}/R_{\text{char}}$ for the axial equation, R_{char} and L_{char} being the mean surface radius and window length of the training set. Without this scaling the physics term falls several orders of magnitude below $\mathcal{L}_{\text{data}}$ and is inert whatever λ ; scaling the axial residual with R_{char} instead of L_{char} has the same effect.

The weights are fixed a priori and shared by all physics-informed families, $\lambda_{\text{phys}} = 0.3$, $\lambda_{\text{bc}} = 0.5$, but λ_{phys} is read as a share of the data gradient: before training, g is set once on the first mini-batch so that the gradient norms of the physics and data terms with respect to the network parameters coincide,

$$g = \|\nabla_{\theta} \mathcal{L}_{\text{data}}\|_2 / \|\nabla_{\theta} \mathcal{L}_{\text{phys}}\|_2. \quad (8)$$

This balancing [41] is what makes strong and weak forms comparable: the weak residual projects the same residual onto oscillating test functions and is damped roughly two-hundred-fold, so that at equal nominal λ the weak term contributed 0.36 % of the data-gradient norm against 37 % for the strong term, and “strong versus weak” degenerated into “physics on versus off”. The measured multipliers in the main stage are $g \approx 0.165$ (0.13–0.22) for the strong and $g \approx 4.97$ (2.8–8.0) for the weak form; g is logged with every PyTorch run.

Weak form (VPINN). Multiplying (3) by $r v_k(r)$ with test functions $v_k = \sin(k\pi r)$, $k = 1, \dots, 8$, $v_k(0) = v_k(1) = 0$, and integrating by parts over $[0, 1]$ gives [42]

$$W_k = - \int_0^1 \hat{\sigma}_{rr} (v_k + r v'_k) dr + \int_0^1 (\hat{\sigma}_{rr} - \hat{\sigma}_{\theta\theta}) v_k dr, \quad (9)$$

evaluated by 32-point Gauss–Legendre quadrature and inserted into (7) with scale $s_W = s_{rr} + s_{\theta\theta}$. No derivative of the network is computed: this, not another way of computing the same derivative, is the substantive difference between the two families.

Training protocol. As noted above, a meaningful comparison between the model families requires that they be brought to the training stage under equal conditions; the protocol described below was designed to that end. The protocol is enforced by construction rather than by configuration: every family is trained by the same trainer class, in which the family selects only the composition of the loss, and one launcher writes every run with its hyperparameters to a single log. It is identical for all families: AdamW [43], learning rate 3×10^{-3} (2×10^{-3} in the extended track), no schedule, weight decay 10^{-4} , gradient-norm clipping at 1.0, mini-batches of 64 cases (32), at most 800 epochs (300), early stopping on the macro-RMSE of an internal early-stopping subset of 15 % of the training cases with patience 40 (30), and restoration of the best state. Every configuration is trained with five seeds (0–4) on the thirteen splits of Table 4 and in the backend comparison, and with three seeds (0–2) in the learning-curve, label-corruption and extended-track stages; the seed governs initialisation, batch order and the internal early-stopping subset, while the test cases of a split are the same for every seed (split seed 42). Learning curves use nested group-aware subsamples of 10/25/50/100 % of the training pool, identical for all families; the subsamples are nested but not stratified by v (the launcher uses `nested_subsamples`, not the stratified variant), so the sparsest level $v = 5$ m/min (122 cases) can be under-represented at 10 %; label corruption adds +250 MPa to $\sigma_{\theta\theta}$ of 0/1/5/10 % of the training cases (0 % being the clean reference; nested lists shared by all families, hold-out clean), the morphology of a solver artefact found in the source rather than Gaussian noise, which averages out over a profile.

Metrics and physical audit. The primary metric is the macro-RMSE (mean over the four components of the component RMSE, MPa). R^2 is reported with a single denominator, the standard deviation of each component over the whole dataset, since a denominator taken from the held-out slice is not comparable between regions. Physical admissibility is audited on the predictions with the operators of the loss (median $|R_1|$ over the profile by finite differences; $|\hat{\sigma}_{rr}|$, $|\hat{\sigma}_{rz}|$ at $r = 1$), always beside the same quantity computed on the FEM data of the same test cases, since the source itself satisfies the traction-free condition only approximately.

Backends. The reference implementation is PyTorch (`torch.autograd`); an independent twin in JAX (`jax.grad`, `optax.adamw`) shares architecture, data, split, the epoch limit and early-stopping rule, and the nominal hyperparameters, but draws initial weights and batch order from its own random stream. The twin implements the strong-form loss with the nominal weight λ_{phys} applied directly: the gradient-norm calibration of (8) is not implemented in it, so the effective physics weight of the two implementations differs. With identical weights copied into both networks, $\partial \hat{\sigma}_{rr} / \partial r$ at 64 random inputs agrees to 8.7×10^{-15} absolute and 9.8×10^{-16} relative (forward pass 1.0×10^{-15}), i.e. to double-precision round-off, so no accuracy difference between backends can originate in automatic differentiation. Each backend is then trained with five seeds on random and `extrap:alpha_max` and compared by the paired test below, with wall-clock training time measured on CPU in double precision. A DeepXDE implementation of the same network obtains its derivatives through `torch.autograd.grad` on the PyTorch backend [37]; it is therefore not a third differentiation engine and is not reported separately.

Statistical treatment. The unit of replication is the training run (seed), not the test case. Within a split, two families are compared by a paired t -test over seeds ($\alpha = 0.05$; no verdict with fewer than three seeds), with $|\Delta|$ reported beside the pooled seed standard deviation as effect size. Ranking of the three families within a split uses Friedman with Nemenyi post hoc, blocks being seeds [44]; indistinguishability is not transitive, so distinguishable pairs are listed rather than “clusters”. Splits are never pooled: ranks in `extrap:alpha_max` and in random describe different tasks. The excess error of an extrapolation split over its matched control is tested by Welch’s unpaired t -test; trends across data fraction and corruption rate by least squares of the seed-wise PINN–MLP gap on $\log(\text{fraction})$ and on the rate. Testing across hold-out cases rather than across seeds was also examined and is not used here: it quantifies the sampling variance of the hold-out, that is, on how many cases one model beats another, and not whether an advantage reproduces when the models are retrained, which is the question at issue.

Fixed architecture and hyperparameters. One architecture and one hyperparameter set were fixed before any held-out region was evaluated and were not tuned per family or per split, so no search could leak a held-out region.

3 Results and Discussion

3.1 FEM results. Verification of the simulation results was carried out by comparison with the established analytical solutions for the drawing stress, those of Siebel and of Avitzur, and by internal consistency tests of the exported fields.

The drawing stress cannot be read from the cases used for the surrogates, whose exported frame is taken after the wire has left the die and the section is unloaded (median $|\int \sigma_{zz} dA|/A$ of 5.3 MPa against an rms σ_{zz} of 167 MPa). It is obtained instead from a dedicated export, in which the reaction at the die reference point, the temperature and the equivalent plastic strain are written as

Table 5 Model families. Within each track all families share the network, the optimiser, the training conditions (epoch limit, early-stopping rule, mini-batch size), the seeds and the splits; only the loss composition differs. R_1 : reduced radial residual (3); R_r, R_z : full residuals (1)–(2); W_k : weak residual (9).

Family	Input	Physics term	BC term	Network derivative
MLP	$(Q, k, \alpha, \mu, v, r)$	–	–	none
PINN (strong)	$(Q, k, \alpha, \mu, v, r)$	R_1 , pointwise	yes	$\partial/\partial r$ (autodiff)
VPINN (weak)	$(Q, k, \alpha, \mu, v, r)$	$W_k, k \leq 8$, 32-pt quadrature	yes	none
mlp2d	$(Q, k, \alpha, \mu, v, z, r)$	–	–	none
pin2d_reduced	$(Q, k, \alpha, \mu, v, z, r)$	R_r without $\partial\tau_{rz}/\partial z$	yes	$\partial/\partial r$
pin2d_full	$(Q, k, \alpha, \mu, v, z, r)$	R_r and R_z , full	yes	$\partial/\partial r, \partial/\partial z$

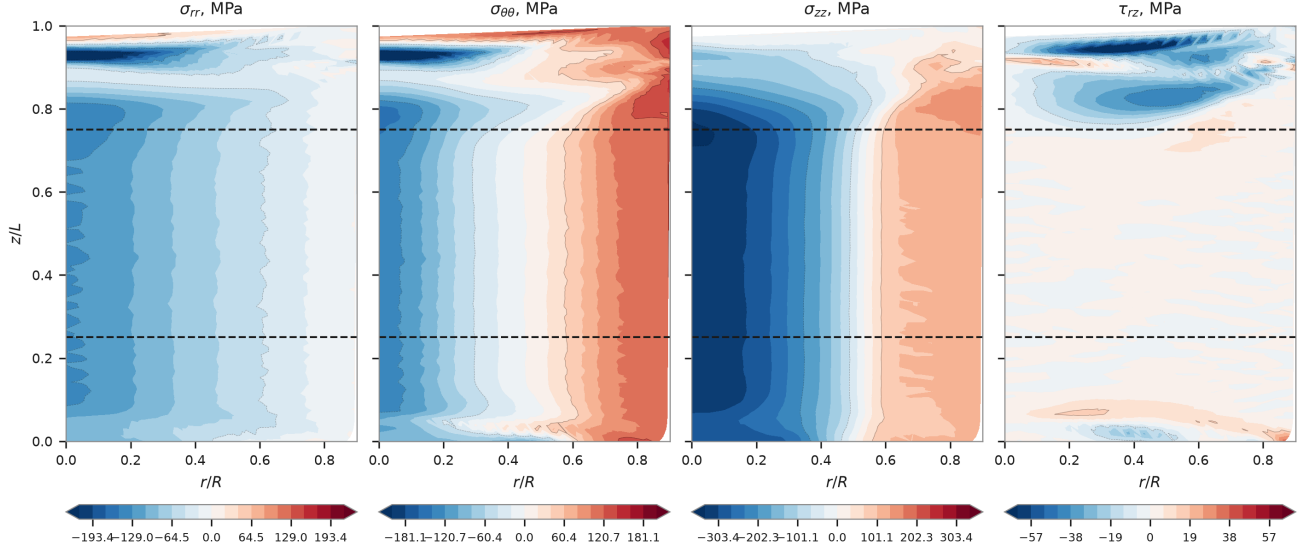


Fig. 4 Distribution of residual stresses obtained from the FEM simulation at $Q = 0.10$ (10% reduction in diameter, 20.1% in area), $k = 0.5$, $\alpha = 12^\circ$, $\mu = 0.05$, $v = 10$ m/min. Panels from left to right: the components σ_{rr} , $\sigma_{\theta\theta}$, σ_{zz} and τ_{rz} over the full axial extent of the wire, each with its own colour scale graduated in MPa. The radius is normalised by the billet radius $R = 18$ mm and the axial coordinate by the model length $L = 139.5$ mm; dashed lines mark the axial window 25–75% of L from which the training data are taken.

histories in time for four runs at a reduction of 1.6% in diameter (3.2% in area) and a land ratio of unity. The axial reaction is averaged over the steady phase of the pass, defined by a die travel of 20 to 100 mm; the signal there is not a plateau but a stick-slip sawtooth whose relative spread is 13.6% at $\mu = 0.025$ and 4.5% at $\mu = 0.1$, so the time average is used, the quantity an energy method estimates. Divided by the area actually reached in the model, 985 mm² at an outlet radius of 17.71 mm, this gives a drawing stress of 54.1, 62.0 and 78.3 MPa at $\alpha = 12^\circ$ for $\mu = 0.025$, 0.05 and 0.1, and 62.3 MPa at $\alpha = 8^\circ$.

Two of the three relations compared below are classical. In the slab-method form of Siebel,

$$\sigma_d = \sigma_f \left[\ln \frac{1}{1-r} (1 + \mu \cot \alpha) + \frac{2}{3} \alpha \right], \quad (10)$$

with $r = 1 - (R_f/R_0)^2$ the reduction in area, α the die semi-angle and σ_f the flow stress, and, in the upper-bound form of Avitzur,

$$\sigma_d = \sigma_f \left[2f(\alpha) \ln \frac{R_0}{R_f} + \frac{2}{\sqrt{3}} \frac{\alpha - \sin \alpha \cos \alpha}{\sin^2 \alpha} + \frac{2}{\sqrt{3}} m \cot \alpha \ln \frac{R_0}{R_f} + \frac{2}{\sqrt{3}} m \right] \quad (11)$$

whose four terms are the ideal deformation work, the redundant shear, the friction on the cone and the friction on the land; $f(\alpha)$ is the closed form of the deformation integral (verified against direct numerical integration to eight digits), $f \rightarrow 1$ as $\alpha \rightarrow 0$, so that the first term reduces to $\ln[1/(1-r)]$, m is the friction factor and L the land length [45–47].

Equation (11) measures the friction through the factor m , the ratio of the interface shear stress to the shear yield stress, whereas the simulations are driven by a Coulomb coefficient μ ; the Siebel relation above already takes μ directly. The conversion $m = \sqrt{3} \mu$ holds only where the normal pressure on the contact equals the flow stress, so it is an assumption about the contact rather than a change of units. Avitzur gives the bound in a second form in which μ enters directly and the land is treated as a bearing along which the axial stress grows under Coulomb friction [46,47]; in the writing of Vega and co-workers [48–50], who build their modification on it, it reads

$$\frac{\sigma_d}{\sigma_f} = \frac{2f(\alpha) \ln \frac{R_0}{R_f} + \frac{2}{\sqrt{3}} \left(\frac{\alpha}{\sin^2 \alpha} - \cot \alpha \right) + 2\mu \left[\cot \alpha \left(1 - \ln \frac{R_0}{R_f} \right) \ln \frac{R_0}{R_f} + \frac{L}{R_f} \right]}{1 + 2\mu \frac{L}{R_f}}, \quad (12)$$

written here for zero back tension and with the flow stress in place of the yield stress of the original. The denominator is not a correction: the axial balance on a cylindrical land, $d\sigma_z/dz = 2\mu p/R_f$ closed by the yield condition $p = \sigma_f - \sigma_z$, integrates to $\sigma_z(L) = \sigma_f - [\sigma_f - \sigma_z(0)] \exp(-2\mu L/R_f)$, of which Eq. (12) is the linearisation; the exact form lies above it by at most 4.0% at the present parameters, so the linearised relation, if anything, understates the bearing term it assumes. Equation (12) therefore carries an explicit assumption about the pressure on the land, $p = \sigma_f - \sigma_z$, which the exported contact data can test.

The modification those authors propose is to drop the assumption that μ is a constant of the lubricant and make it a function of the temperature measured by a thermocouple set in the die above the bearing, which they take to be proportional to the temperature of the wire–die interface,

$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^{m_\mu}, \quad (13)$$

with μ_0 the coefficient at a reference temperature T_0 ; inverting Eq. (12) against their own force measurements they obtain $\mu_0 = 0.0125$, $m_\mu = 2.68$ and $T_0 = 15.6^\circ\text{C}$ for copper drawn in oil at 1–7 m/s, the temperatures entering the ratio in degrees Celsius. In the present simulations μ is prescribed and constant, so Eq. (13) has no counterpart in the model; what it contributes to the comparison is stated below.

All three relations are evaluated with the same representative flow stress derived from the constitutive equation (Section 2.2), 300–305 MPa for these runs, and with no fitted parameter, so that the columns of Table 6 differ only in the form of their work terms; the flow stress enters as a multiplier and shifts every prediction together.

Both Avitur columns lie above the computed drawing stress at every point, as the upper-bound character of the underlying solution leads one to expect, although the theorem is not strictly available here, since the model is elastic–plastic with hardening and thermal softening and Eq. (12) is in addition a linearisation. The margins are not small, and they order themselves by how each relation treats the land: 36.2 % on average in the Coulomb form and 55.2 % in the friction-factor form. Siebel is the closest, at 17.3 %, and it is the only one of the three with no land term at all; it is also the only one that falls below the computed value, which it does at every point, and its shortfall grows monotonically with friction, from 1.7 % at $\mu = 0.025$ to 26.7 % at $\mu = 0.1$. Its low mean deviation is therefore a cancellation at the lowest-friction run, not a uniform agreement. At a land ratio of unity the land is 89–93 % of the contact area, so this is where the comparison is decided.

The exported contact reaction locates the error. Through its yield closure $p = \sigma_f - \sigma_z$, Eq. (12) implies a pressure of 240–251 MPa at the entrance to the land, falling to a mean of 221–237 MPa along it, whereas the mean contact pressure in the model is 122 MPa: a factor of 1.97 to 2.06 on the entrance value and 1.80 to 1.94 on the mean. The friction-factor form is further out still, since $m = \sqrt{3}\mu$ is equivalent to taking the pressure on the land equal to the flow stress, 300–305 MPa, a factor of 2.5. The sensitivity to friction confirms the diagnosis independently of the magnitudes: from $\mu = 0.025$ to $\mu = 0.1$ at $\alpha = 12^\circ$ the computed drawing stress rises by 323 MPa per unit μ , against 439 for Eq. (12), 670 for Eq. (11) and 57 for Siebel, whose insensitivity to friction is the other face of its missing land term. Substituting the computed contact pressure into the bearing term of Eq. (12) and changing nothing else gives 66.1, 73.1, 88.8 and 73.3 MPa for the four runs: a mean deviation of 17.8 % instead of 36.2 %, and a slope of 303 MPa per unit μ , 6.1 % below the 322.7 MPa of the model. The exports say directly that the yield closure does not hold there: on the land the measured pressure and the axial stress sum to 176–200 MPa against a flow stress of 300–305 MPa, so the material on the bearing is not at yield, and the two relations that assume it is overstate the only term that carries most of their answer. That, and not the calibration of the flow stress, is what both bearing models get wrong. The Coulomb branch itself remains admissible throughout: μp reaches 25 MPa at most even at the pressure the relation assumes, and 12 MPa at the computed pressure, against a shear yield stress of 173–176 MPa, so nowhere does the contact approach sticking.

A second departure, of a different kind, survives that correction at the lowest friction. Setting $\mu = 0$ in Eq. (12) leaves 58.8 MPa at $\alpha = 12^\circ$, already 8.8 % above the 54.1 MPa the model returns with friction, so no treatment of the contact can close that row at the flow stress used here. The offset lies in the redundant-shear term,

which carries 29 to 65 % of the predicted drawing stress against 8 to 13 % for the ideal work. The inhomogeneity parameter is 17.1 at $\alpha = 8^\circ$ and 25.6 at $\alpha = 12^\circ$, against 1.81 in the case for which Eq. (12) was calibrated, where the ideal work carries 74 % of the frictionless drawing stress. The two failures are therefore distinct: the bearing term is wrong about the contact, and the shear term is being used far outside the geometry in which the assumed velocity field is a good approximation.

Equation (13) cannot be tested on this campaign and must not be carried over with its published constants. The Coulomb coefficient of the simulations is an input and does not respond to the 15–21 °C mean rise the model produces; and reading μ_0 as the prescribed coefficient at the initial 20 °C would multiply it by 2.3 to 3.1 at the measured pass-mean temperature of 27.5–30.6 °C, or by 4.6 to 6.1 at the reference temperature of the original calibration, in either case moving every prediction further from the computed value rather than closer. What the relation contributes here is the statement of what a friction model for this process would have to contain, and the identification of what a campaign in which μ is prescribed rather than observed cannot supply.

The comparison can be closed completely, with no quantity taken from the finite-element model. The homogeneous strain is $2 \ln(R_0/R_f)$ and the total strain follows from an excess-work factor $\Phi = 0.8 + \Delta/4.4$; the strain rate is obtained from the time to traverse the cone at constant volume flow, $\dot{\epsilon} = 6v R_f^2 \tan \alpha \ln(R_0/R_f)/(R_0^3 - R_f^3)$; and the flow stress is the work-weighted mean of the constitutive equation over that path:

$$\epsilon_{\text{hom}} = 2 \ln \frac{R_0}{R_f}, \quad \Phi = 0.8 + \frac{\Delta}{4.4}, \quad \epsilon_{\text{eff}} = \Phi \epsilon_{\text{hom}}, \quad (14)$$

$$\dot{\epsilon} = \frac{6v R_f^2 \tan \alpha \ln(R_0/R_f)}{R_0^3 - R_f^3}, \quad \langle \sigma_f \rangle = A + \frac{B \epsilon_{\text{eff}}^n}{n+1}, \quad \Delta T = \frac{\beta \sigma_d}{\rho c}, \quad (15)$$

where Δ is the inhomogeneity parameter of the deformation zone and $\langle \sigma_f \rangle$ the plastic-work-weighted mean of the strain term of the constitutive equation, taken before its rate and temperature multipliers [51]. The excess-work correlation is quoted in the literature in more than one form; replacing $\Phi = 0.8 + \Delta/4.4$ by the alternative $\Phi = 0.88 + 0.12 \Delta$ changes Φ by a factor of 1.6 to 1.7 but the flow stress, and the estimate below, by only 1.3 to 1.6 %, because at $\epsilon_{\text{hom}} = 0.032$ the hardening term of the constitutive equation is flat. The choice therefore does not carry the comparison. The temperature does not need a separate source, because it closes on the deformation work itself: that work per unit volume is the part of σ_d raised in the cone, a fraction β of which becomes heat, so the mean rise is $\Delta T = \beta \sigma_d/(\rho c)$; since σ_d depends on the flow stress and the flow stress on the temperature, this is a fixed point, reached in five iterations. The friction on the land is left out of this balance deliberately: its heat is released in a thin surface layer and is shared with the die, so it does not raise the mean temperature of the section, whereas the friction on the cone, 2 to 14 % of the estimate, is retained because it is dissipated in the deforming material itself. Carrying the land term into the balance as well would raise the estimated rise by 25 to 120 % and break the agreement with the measured values of Table 8. For the four runs the fixed point returns a flow stress of 330–338 MPa. This is 10 % above the 300–305 MPa used in Table 6, and the difference is not a disagreement: the fixed point evaluates the constitutive equation once, at $\epsilon_{\text{eff}} = 0.215$ and 0.152, $\dot{\epsilon} = 15.6$ and 10.3 s^{−1} and the pass-mean temperature, whereas the representative value of Section 2.2 averages the same equation along the deformation path of every material point, most of which lies at lower strain. The closed estimate is quoted with its own value throughout, and no column of Table 6 uses it.

The temperature closure can be checked directly, since the dedicated export also carries the temperature history. The self-

Table 6 Drawing stress (MPa) for the four runs: the finite-element value against the Siebel relation, Eq. (10), the Avitzur upper bound in its friction-factor form, Eq. (11) with $m = \sqrt{3}\mu$, and the modified Avitzur relation in the Coulomb form of Eq. (12). All three use the flow stress derived from the constitutive equation, 300–305 MPa, and no fitted parameter. Reduction Q is nominal, in diameter; the achieved outlet radius is 17.71 mm in every run. The last two rows are the mean absolute deviation from the finite-element value and the secant slope $d\sigma_d/d\mu$ between the $\mu = 0.025$ and $\mu = 0.1$ rows at $\alpha = 12^\circ$, for which the finite-element value is 323 MPa.

Parameters					Analytical			Simulation
Q	α	v , m/min	k	μ	Siebel	Avitzur	Avitzur_mod	FEM
0.015	12°	40	1.0	0.025	53.2	75.1	71.5	54.1
0.015	12°	40	1.0	0.050	54.2	91.1	82.8	62.0
0.015	12°	40	1.0	0.100	57.4	125.4	104.4	78.3
0.015	8°	40	1.0	0.100	44.7	109.0	90.8	62.3
mean absolute deviation					17.3 %	55.2 %	36.2 %	—
$d\sigma_d/d\mu$ at $\alpha = 12^\circ$, MPa					57	670	439	323

Table 7 Intermediate quantities of the closed self-consistent estimate, Eqs. (14)–(15), for the four runs: inhomogeneity parameter Δ , Backofen excess-work factor Φ , effective strain ϵ_{eff} , mean strain rate $\dot{\epsilon}$, flow stress σ_f from the constitutive equation at the fixed-point temperature, and the deformation part of the drawing stress, σ_d^{def} , being the ideal work, the redundant shear and the friction on the cone, with the friction on the land excluded for the reason given in the text. Nothing is taken from the finite-element model. The nominal geometry is used throughout, $\epsilon_{\text{hom}} = 0.032$ at $R_f = 17.71$ mm, so Δ here is the value for that geometry; evaluated instead at the outlet radius measured in each run it is 25.2–25.7 and 17.0.

α	μ	Δ	Φ	ϵ_{eff}	$\dot{\epsilon}$, s ⁻¹	σ_f , MPa	σ_d^{def} , MPa
12°	0.025	25.6	6.62	0.215	15.6	338.2	67.1
12°	0.050	25.6	6.62	0.215	15.6	338.1	68.4
12°	0.100	25.6	6.62	0.215	15.6	337.9	70.9
8°	0.100	17.1	4.70	0.152	10.3	330.1	53.9

consistent estimate reproduces the cross-section-averaged rise measured in the model to within 12 % on average, and to within 19 % on every run (Table 8); the heating can therefore be taken from the deformation work itself, with this comparison as its justification, rather than from an assumed thermal boundary condition. The closure is not exact in its friction dependence: over the friction range it moves the estimated rise by 1.0 °C against 6.3 °C measured, so some of the land-friction heat does reach the section, but including that term in full would overshoot the measured rise by more than it now falls short. Temperatures are stated in degrees Celsius throughout, the scale in which the constitutive model is defined, with a transition temperature of 20 °C equal to the initial temperature of every run; a rise in degrees Celsius and a rise in kelvin are the same number, but the absolute values are not, and Eq. (13) is calibrated on the Celsius scale.

Table 8 Mean temperature rise during the pass and the resulting pass-mean temperature: the self-consistent analytical estimate, in which the deformation work and the temperature close on each other and nothing is read from the model, against the cross-section average of the exported temperature history. Every run starts at 20 °C, and the pass-mean temperature is formed from the measured rise. The peak rise, reached at the surface, is 45.7–74.8 °C, three to four times the section mean. Reduction 1.6 % in diameter (3.2 % in area), land ratio unity.

α	μ	ΔT analytical, °C	ΔT measured, °C	T pass mean, °C
12°	0.025	16.3	15.0	27.5
12°	0.050	16.6	16.9	28.4
12°	0.100	17.3	21.3	30.6
8°	0.100	13.1	15.8	27.9

Three properties of the exported histories support the extraction independently of the analytical comparison. The friction coefficient is recovered from the data rather than read from the file name: over a die travel of 115 to 140 mm the cone has left the

wire and only the cylindrical land remains in contact, the contact normal is then strictly radial, and the ratio of the axial to the radial reaction returns 0.0250, 0.0500, 0.1000 and 0.0999 for the four runs. The mean contact pressure is 122 MPa and varies by 1 MPa over the four runs, with neither the semi-angle nor the friction, as it must when a cylindrical land is 89–93 % of the contact area. And the flow stress recovered from the heat alone, over the inner half of the radius where friction contributes none, is 292–329 MPa, in agreement with the 300–305 MPa derived from the constitutive equation; beyond that radius the same estimate runs up past 500 MPa as the friction heat of the land reaches it, which is itself a check that the core is the right place to take it.

The region of the design covered by these runs limits what the comparison establishes. All four lie at a reduction of 1.6 % with a land as long as the outlet radius, a combination in which the inhomogeneity parameter is 25.6 at $\alpha = 12^\circ$ and 17.1 at $\alpha = 8^\circ$, against the range of 1 to 3 in which these solutions are normally applied and against which they have been validated; they remain valid outside it but become progressively loose, because the assumed velocity field departs further from the actual flow. The ideal work accounts for 8 to 13 % of the predicted drawing stress and the total exceeds it by a factor of 7.7 to 12.4. The deviations of Table 6 therefore characterise the closed-form solutions in this corner of the design, not the accuracy of the finite-element model, and they do not transfer to the industrial range of reductions, where the ideal work dominates and the land is a small fraction of the contact. What does transfer is the bearing diagnosis: wherever the land is long relative to the reduction, a closed-form solution that puts it at the yield pressure will over-predict the drawing force by about as much as that pressure is overstated.

In addition, five internal consistency tests were run on the 1756 retained cases. The exported cross-section is self-equilibrated to 3.2 %: the median $|\int \sigma_{zz} dA|/A$ is 5.3 MPa against an rms σ_{zz} of 167 MPa. Local equilibrium closes to 2.1 % of the sum of the moduli of its terms. At the free surface $|\sigma_{rr}|$ is 4.0 MPa against 54.0 MPa in the bulk, i.e. 7.4 % (a nodal value; the 3.52 MPa quoted below is the median of the outermost bin of the 20-bin

profile), on the axis $|\sigma_{rr} - \sigma_{\theta\theta}| = 1.07$ MPa (0.6 %), and the residual field stays below the flow stress everywhere: the maximum von Mises stress over the retained cases is 232.6 MPa, against the 300–305 MPa that the constitutive equation returns for the deformation histories of this pass (Section 2.2). Kinematically, the exit radius follows $R_0(1 - Q)$ to 0.11 % (ratio 1.0011) and $|\text{trLE}| = 5.1 \times 10^{-4}$, i.e. plastic incompressibility holds to 0.05 %. These tests check the mesh, the contact and the solver of this model, not its correspondence to real wire.

3.2 ML models results. All seed-replicated results below were obtained on the cleaned dataset of 1756 cases (Section 2.2) in 341 training runs: 195 for the split matrix, 72 for the learning curves, 36 for label corruption, 20 for the backend comparison and 18 for the two-dimensional ablation; the one exception, the single-run comparison of Table 11, is described in Section 3.2.3. The three one-dimensional families – the data-driven MLP, the strong-form PINN and the weak-form VPINN – share the network, the optimiser, the epoch limit, the early-stopping rule, the mini-batch size and the seeds, and differ only in the composition of the loss (Section 2.3). Accuracy is reported as macro-RMSE, the mean of the four component RMSEs in MPa, as mean $\pm$ standard deviation over five seeds; differences between families are tested with a paired t -test over seeds [44] and called established at $p < 0.05$. Because τ_{rz} is two orders of magnitude smaller than the normal components, a coefficient of determination over the normal components only, R_n^2 , with a single denominator (global denominator: the variance of each component over the whole dataset) is given alongside; where a per-component R^2 or NRMSE below uses the held-out slice as denominator, this is stated at the number. Throughout this section the finite-element solution serves as the reference field, and this sets the domain of applicability of the results reported below: they characterise the surrogate models with respect to that reference – accuracy on it, transferability beyond the training range and robustness to defects of the source – and are stated in these terms (see the closing paragraph of this section).

3.2.1 Extrapolation versus interpolation. Table 13 gives the macro-RMSE on all thirteen splits. On the random hold-out all three families reach 8.0 MPa with $R_n^2 = 0.99$ (global denominator), against 66 MPa for the constant predictor, and are indistinguishable. The informative contrast is between each region split and its matched control of the same size, which separates the cost of leaving the training range from the cost of having fewer training cases.

The cost of extrapolation depends on the axis by almost an order of magnitude (Tables 13 and 9). Holding out the longest die land ($k = 1.0$) costs practically nothing: the excess over the control is 1.13 ± 0.58 MPa for the MLP (established by Welch’s test, Section 2.3) and 0.11 ± 0.40 MPa for the PINN (not established). Holding out the largest semi-angle costs a factor of 1.6 with R_n^2 (global denominator) still at 0.96, the largest reduction a factor of 3.2 ($R_n^2 = 0.78$ –0.83), the highest speed a factor of 5.9. The spread between axes is far larger than any difference between families, so a single “extrapolation error” would be uninformative.

Removing an interior level is not free either. Along α , where the volumes are nearly equal, withholding the interior level $\alpha = 12^\circ$ gives 8.7–9.1 MPa ($R_n^2 = 0.98$, global denominator) and withholding the extreme level $\alpha = 20^\circ$ gives 13.0–13.6 MPa ($R_n^2 = 0.96$ –0.97), against 8.0 MPa on the random hold-out: most of the price is paid for the hole in a five-level factor grid, and the direction of the hole adds comparatively little. Along Q the same pattern is steeper, 11.0–12.4 MPa for $Q = 0.10$ against 23.4–26.3 MPa for $Q = 0.20$.

The speed axis shows the largest loss of accuracy, and its interpretation requires care. In the source simulations each level of v coincides one-to-one with a separate batch of runs (Section 2.2), so withholding the highest level removes not only the extreme level of a factor but an entire batch: extrapolation along v is here

inseparable from a transfer to a batch that is not represented in training. With $v = 250$ m/min withheld the error is 48–56 MPa, still below the 71 MPa of the constant predictor, but R_n^2 (global denominator) falls to 0.23–0.45, the shear component is reproduced no better than by a constant (NRMSE 1.6–1.9, held-out standard deviation as denominator) and the MLP becomes unstable across seeds (± 19 MPa). These figures thus measure the joint effect of leaving the range of v and of changing the batch of runs, and the loss of accuracy on this split cannot be attributed to extrapolation along the speed axis alone; no claim about the physical effect of speed is made from it.

Table 14 summarises the seed-paired tests. The PINN has a lower macro-RMSE than the MLP on all thirteen splits, and the difference is established on four: the semi-angle extrapolation ($\Delta = -0.58$ MPa, $p = 0.040$), the land-length extrapolation (-1.16 MPa, $p = 0.033$), the joint departure from the range in semi-angle and reduction (-1.06 MPa, $p = 0.011$) and the matched control of the speed split (-0.29 MPa, $p = 0.006$). With a Holm step-down correction [44,52] over the thirteen splits none of the pairwise differences remains significant (smallest adjusted $p = 0.079$ for PINN–MLP, 0.195 for VPINN–MLP); the four verdicts are read as a consistent direction, not as isolated proofs. The VPINN is established better than the MLP on two splits (interp:Q_mid, $p = 0.015$; matched:v_max, $p = 0.022$). The strong form is established better than the weak form on extrapol:k_max (-0.70 MPa, $p = 0.018$) and, by a margin smaller than the seed spread, on matched:v_max (-0.11 MPa, $p = 0.046$); on the remaining eleven splits the two physics-informed forms are indistinguishable. A Friedman test over seeds [44,53] rejects equality on three of the four splits where the paired test does (extrap:alpha_max, $p = 0.041$; extrap:k_max, $p = 0.022$; matched:v_max, $p = 0.007$; on the corner $p = 0.091$), with the Nemenyi post-hoc separating only the MLP–PINN pair; since indistinguishability there is not transitive, no “single cluster” is claimed. In absolute terms the advantage is small, 0.3–1.9 MPa at an error level of 8–16 MPa and of the same order as the seed spread; accuracy alone does not justify the physics term (Section 3.2.3).

The learning curves of Table 10 do not support the expectation that a physics prior pays off mainly when data are scarce. On the random split the PINN–MLP gap shrinks with more data ($-0.61 \rightarrow -0.10$ MPa), as that expectation predicts; on the semi-angle extrapolation it moves the other way ($+0.24 \rightarrow -0.54$ MPa). Neither trend is established (slopes $+0.20$ MPa per log-unit, $p = 0.21$, and -0.35 MPa, $p = 0.070$); the two isolated significant points ($p = 0.048$ at 50 %, $p = 0.033$ at 25 %) have opposite signs, as eight comparisons produce by chance, and at the smallest fraction the gap is not established on either split. The data-scarcity claim is therefore not made.

3.2.2 Backend comparison. The strong-form PINN was trained in two independent implementations of the same protocol, in PyTorch and in JAX, on the random split and the semi-angle extrapolation, five seeds each. The engines were checked first: on bit-identical weights and inputs the derivative $\partial\sigma_{rr}/\partial r$ from torch.autograd and from jax.grad agrees to 8.7×10^{-15} absolute (9.8×10^{-16} relative), i.e. to machine precision. A DeepXDE implementation [37] of the same network runs on the PyTorch backend and differentiates with the same torch.autograd.grad, so the comparison involves two automatic-differentiation engines, not three.

The implementations are statistically indistinguishable in accuracy (Table 15): $+0.14$ MPa on the random split ($p = 0.27$) and $+0.43$ MPa on the extrapolation ($p = 0.17$), with the same sign on both. The second gap is of the order of the seed spread: a difference of this size arises between two implementations of one method that differ in their random streams and, here, in the effective physics weight. Differences of a few tenths of a megapascal between backends are of the same order as the spread between two implementations of one method and cannot be attributed to the

Table 9 Cost of holding out a whole factor level, averaged over the three families: macro-RMSE on the held-out region, on the matched control of the same size, their ratio, R_n^2 (global denominator) on the held-out region, and the constant predictor (MPa).

Held-out region	Outside	Control	Ratio	R_n^2 outside	Constant
$k = 1.0$	8.9	8.3	$\times 1.08$	+0.98	67
$\alpha = 20^\circ$	13.2	8.1	$\times 1.64$	+0.96	61
$\alpha = 20^\circ, Q \geq 0.15$	16.1	7.3	$\times 2.20$	+0.94	61
$Q = 0.20$	24.7	7.8	$\times 3.15$	+0.81	60
$v = 250$ m/min	50.9	8.6	$\times 5.93$	+0.36	71

Table 10 Learning curves: macro-RMSE (MPa) versus the fraction of the training pool, three seeds. Δ is PINN minus MLP with the seed-paired p -value. Slope of Δ on $\log(\text{fraction})$: +0.20 MPa ($p = 0.21$) on random, -0.35 MPa ($p = 0.070$) on extrap:alpha_max.

Split	Fraction	n_{train}	MLP	PINN	VPINN	Δ	p
random	10 %	149	13.09 $\pm$ 0.67	12.48 $\pm$ 0.18	12.98 $\pm$ 0.10	-0.61	0.317
	25 %	373	9.92 $\pm$ 0.40	9.46 $\pm$ 0.39	9.92 $\pm$ 0.18	-0.46	0.099
	50 %	746	8.80 $\pm$ 0.15	8.34 $\pm$ 0.06	8.53 $\pm$ 0.21	-0.46	0.048
	100 %	1493	8.19 $\pm$ 0.34	8.09 $\pm$ 0.20	8.03 $\pm$ 0.19	-0.10	0.718
extrap:alpha_max	10 %	138	20.48 $\pm$ 0.63	20.72 $\pm$ 0.21	20.68 $\pm$ 0.67	+0.24	0.498
	25 %	344	16.74 $\pm$ 0.99	17.33 $\pm$ 1.17	17.03 $\pm$ 0.33	+0.59	0.033
	50 %	688	14.01 $\pm$ 0.40	14.06 $\pm$ 0.70	14.17 $\pm$ 0.95	+0.05	0.886
	100 %	1377	13.48 $\pm$ 0.58	12.95 $\pm$ 0.36	12.70 $\pm$ 0.48	-0.54	0.238

differentiation engine, so no conclusion about a preferred backend follows from accuracy. The ranking holds for CPU execution in double precision and must not be transferred elsewhere: on a GPU in single precision the just-in-time compilation of JAX is amortised over long uniform runs, whereas here early stopping cuts the runs short.

3.2.3 PINN, MLP and FEM. On individual components the families are as close as on the macro figure: on the random split of the one-dimensional track the component RMSE (five seeds) is 5.0–5.1 MPa for σ_{rr} , 7.0 MPa for $\sigma_{\theta\theta}$, 19.5–19.7 MPa for σ_{zz} and 0.36–0.37 MPa for τ_{rz} for all three families; the low R^2 of the shear component (0.76–0.78 against 0.98–0.99 for the normal components; per-component R^2 with the held-out standard deviation as denominator) reflects the axial collapse of the profile, under which the mean absolute value of τ_{rz} falls by 79 % because the component changes sign along z (Section 2.2). The comparison with the finite-element source is made on the extended track, where z is an input and the full 8×20 field is predicted (Section 2.3), because only there can both equilibrium equations be evaluated on the prediction.

This comparison uses an earlier single-seed training on 1498 cases with a common hold-out of 261 cases: the 1848-case set cleaned by the window mask alone (89 cases removed, 1759 in total), on whose random split of 1571/277 the mask removes 73 training and 16 test cases. The 1756-case set of Section 2.2 removes in addition the three runs whose undrawn tail lies outside the window; the other stages of the matrix, including the seed-replicated ablation of Table 12, were run on it. The dimensionless residuals of Table 11 and Fig. 6 are the equilibrium residuals R_r , R_z of Section 2.3 divided by the scales of the loss term (7), $s_{R_1} = (s_{rr} + s_{\theta\theta})/R_{\text{char}}$ and $s_{R_z} = s_{zz}/L_{\text{char}} + s_{rz}/R_{\text{char}}$, computed once from the FEM data and applied identically to all families, since otherwise each family would be normalised by itself and the rows would not be comparable; a value of 0.1 means that the equation fails to close by 10 % of the characteristic magnitude of its own terms. Medians are taken over the interior grid nodes, where the central difference is second-order accurate; at the edges of the grid it degenerates to a one-sided difference. The FEM row is the baseline against which every admissibility figure is read.

Table 11 shows where the families part. In accuracy they are equivalent to within 0.35 MPa of macro-RMSE, and their fields

are visually alike (Fig. 5, the case with the median error of the full PINN rather than the best). Within each component the colour scale is shared by the FEM reference and all families, so that the panels of one column are directly comparable; in Figs. 5 and 6 r is normalised by the surface radius of the same cross-section and z by the window length, and the radial axis is stretched ($L/R \approx 3.5$). In admissibility they are not (Fig. 6). The boundary quantity is evaluated on the material surface, at $r/R = 1$, on the four components of the stress tensor. The MLP violates radial equilibrium 2.8 times more strongly than the finite-element source it was trained on, both PINNs about 1.5 times; a residual of 0.115 means that the radial equation fails to close by 11.5 % of the characteristic magnitude of its own terms. The axial equation is improved only by the full form, the one configuration whose loss contains R_z : 0.047 against 0.091 for the source and 0.074 for the MLP. The separation is sharpest at the free surface: the MLP satisfies the traction-free condition at the level of the source (2.90 against 3.54 MPa), whereas the reduced and full PINNs satisfy it 4.5 and 6.3 times more strictly than the data they were trained on. Since the finite-element field is affected by nodal extrapolation and averaging, a surrogate that violates equilibrium and the boundary condition by less than its source is, in this specific sense, more physically admissible than the data.

The ordering is the same with seed statistics on the one-dimensional track. On the random split the median reduced radial residual is 402 MPa/m for the source, 1351 ± 146 for the MLP, 749 ± 42 for the PINN and 1071 ± 26 for the VPINN, and $|\sigma_{rr}(r=1)|$ is 3.48 , 3.41 ± 0.61 , 0.87 ± 0.21 and 0.64 ± 0.14 MPa respectively. The advantage persists with $\alpha = 20^\circ$ withheld (3.95 ± 0.82 MPa for the MLP against 1.74 ± 0.32 and 1.80 ± 0.66 , source 3.35) and disappears only on the speed extrapolation discussed above, where all three families lose accuracy (28–34 MPa for every family).

The ablation in Table 12 isolates the axial equation; its residuals are evaluated by the training code in MPa/m on the 1756-case splits and are not numerically comparable to the dimensionless medians of Table 11. Both physics-informed forms reduce the radial residual by 40–52 %, since both contain the radial equation. The axial residual falls by 33–41 % with the full form and by only 9–14 % with the reduced one, which drops $\partial\sigma_{zz}/\partial z$, a term amounting to 68 % of the retained ones, and therefore cannot enforce axial balance. The gain in axial equilibrium thus comes from the axial term in the loss, not from “physics in general”, and costs no accuracy:

Table 11 Accuracy and physical admissibility of the three two-dimensional families on a common hold-out of 261 cases. Accuracy: component RMSE (MPa) / R^2 and macro-RMSE; admissibility: the dimensionless equilibrium residuals $|R_r|$, $|R_z|$ and the traction-free violation $|\sigma_{rr}(r=1)|$, MPa. Definitions, denominators and the training pool are given in the text.

	Accuracy, RMSE (MPa) / R^2					Admissibility		
	σ_{rr}	$\sigma_{\theta\theta}$	σ_{zz}	τ_{rz}	macro	$ R_r $	$ R_z $	$ \sigma_{rr}(r=1) $, MPa
FEM (source)	—	—	—	—	—	0.041	0.091	3.54
MLP	3.52/0.994	5.87/0.993	19.45/0.984	1.25/0.934	10.33	0.115	0.074	2.90
PINN, reduced form	3.69/0.993	6.04/0.993	20.12/0.983	1.27/0.932	10.68	0.064	0.064	0.78
PINN, full form	3.53/0.994	5.97/0.993	20.15/0.982	1.25/0.935	10.67	0.062	0.047	0.56

Table 12 Ablation on the extended track (z as input, 8×20 grid), three seeds: macro-RMSE (MPa) and median full-form equilibrium residuals (MPa/m) on the held-out cases; in brackets, change relative to the MLP with the same input. The three configurations differ only in the physics terms of the loss.

Split	Configuration	Macro-RMSE	$ R_r $	$ R_z $
random	MLP, z as input	7.81 ± 0.22	1040	238
	PINN, reduced form	8.05 ± 0.44	626 (−40 %)	217 (−9 %)
	PINN, full form	8.02 ± 0.56	553 (−47 %)	140 (−41 %)
extrap:alpha_max	MLP, z as input	12.70 ± 0.03	1426	281
	PINN, reduced form	12.18 ± 0.34	739 (−48 %)	242 (−14 %)
	PINN, full form	12.21 ± 0.51	686 (−52 %)	188 (−33 %)

the macro-RMSE changes by +0.2 MPa on the random split and −0.5 MPa on the extrapolation, within the seed spread.

The corruption experiment of Table 16 yields the strongest and most interpretable effect of the physics term. On clean labels the PINN and the MLP are indistinguishable ($\Delta = -0.10$ MPa, $p = 0.72$): the term adds nothing where data and equations already agree. As labels are corrupted the gap opens monotonically, $-0.10 \rightarrow -0.48 \rightarrow -1.11 \rightarrow -1.59$ MPa, and is established at 5 and 10%. A regression of the gap on the corrupted fraction over the twelve seed-level points gives −14.2 MPa per unit fraction ($p = 0.00036$, $R^2 = 0.74$): the advantage grows by about 1.4 MPa for every 10 % of corrupted labels. At 10 % the MLP has degraded by a factor of 1.81 and the PINN by 1.64; the weak form (1.79) is not protected in the same way. The mechanism is the one seen in Table 11: the strong-form residual and the traction-free term pull the solution towards an admissible field, and the pull matters exactly when the labels contradict it. Taken together, Tables 11 and 16 locate the benefit of the physics term not in accuracy against a clean finite-element field, where it is marginal, but in the admissibility of the solution and its resistance to defects of the data source.

Limitations.. The surrogates are trained and evaluated against a single finite-element model that has not been validated against physical measurements (Section 2.2). This study therefore does not assess how accurately any of the models represents the wire-drawing process; it assesses properties of the surrogates relative to their data source, transferability beyond the training range and robustness to defects of that source, which are measurable without experimental validation and are what is claimed. The factorial grid is incomplete and unbalanced: the level $v = 5$ m/min is represented by 122 cases against 554–607 for the other levels, speed coincides with the simulation batch, and 13.2 % of the raw cases are bit-identical duplicates of another case differing only in the friction label, so friction cannot be used as an extrapolation axis. The shear component τ_{rz} is essentially not predicted from the one-dimensional representation: its NRMSE (held-out standard deviation as denominator) is 0.47–0.49 on the random hold-out and 1.6–1.9 with the highest speed withheld, against a physical scale of 0.67 MPa mean absolute value, two orders of magnitude below the normal components; macro-averaged figures are therefore dominated by the normal components, and per-component figures are given alongside them.

Table 13 Macro-averaged RMSE on the held-out cases for the 13 splits of the main track, MPa, mean $\pm$ sd over 5 seeds. Split types are defined in Section 2.2.

Split	Held-out region	n_{train}	n_{test}
random	random 15 % of all cases	1493	263
interp:Q_mid	$Q = 0.10$ (interior level)	1438	318
interp:alpha_mid	$\alpha = 12^\circ$ (interior level)	1416	340
extrap:Q_high	$Q = 0.20$ (highest level)	1484	272
extrap:k_max	$k = 1.0$ (highest level)	1466	290
extrap:alpha_max	$\alpha = 20^\circ$ (highest level)	1377	379
extrap:v_max	$v = 250$ m/min (highest level)	1255	501
extrap_joint:corner	$\alpha = 20^\circ$ and $Q \geq 0.15$	1642	114
matched:Q_high	random, n_{test} of extrap:Q_high	1484	272
matched:k_max	random, n_{test} of extrap:k_max	1466	290
matched:alpha_max	random, n_{test} of extrap:alpha_max	1377	379
matched:v_max	random, n_{test} of extrap:v_max	1255	501
matched:corner	random, n_{test} of extrap_joint:corner	1642	114

4 Conclusions

In the present work a series of axisymmetric finite-element simulations of cold drawing of AISI 1020 wire was carried out over a factorial grid of process parameters, and the computed residual-stress fields were assembled and filtered into a working dataset of 1756 cases (Section 2.2). On this dataset three families of surrogate models were prepared – a data-driven multilayer perceptron, a strong-form physics-informed network and its weak, variational counterpart – sharing the architecture, the optimiser, the epoch limit, the early-stopping rule, the mini-batch size and the seeds and differing only in the composition of the loss (Section 2.3). For the comparative assessment several partitions of the preprocessed sample were constructed: a random hold-out, the removal of an interior level of a factor, the removal of the highest level of a single factor, the removal as a whole of a region of the joint space of two factors – the die semi-angle and the reduction – left uncovered by the design, the held-out cases of that region still lying inside the training range along each of the two axes taken separately, and random controls of the same size as the corresponding held-out region, which separate the cost of leaving the training range from the cost of a smaller training pool (Section 3.2). Auxiliary series addressed the size of the training pool, the software backend, the

Table 14 Seed-paired comparison of the physics-informed families with the data-driven baseline on every split of Table 13. Δ is the mean over seeds of the per-seed difference in macro-RMSE, MPa; negative favours the physics-informed model. Two-sided paired t -test, $n = 5$ pairs, $\alpha = 0.05$. After a Holm correction over the 13 splits, 0 PINN – MLP and 0 VPINN – MLP differences remain significant.

Split	PINN – MLP				Δ , MPa
	Δ , MPa	t	p	verdict	
random	−0.07	−0.47	0.664	n.s.	−0.06
interp:Q_mid	−0.20	−0.50	0.643	n.s.	−1.44
interp:alpha_mid	−0.37	−2.02	0.113	n.s.	−0.07
extrap:Q_high	−1.92	−2.58	0.062	n.s.	−2.87
extrap:k_max	−1.16	−3.18	0.033	PINN better	−0.46
extrap:alpha_max	−0.58	−3.01	0.040	PINN better	−0.66
extrap:v_max	−7.48	−0.73	0.508	n.s.	−7.41
extrap_joint:corner	−1.06	−4.42	0.011	PINN better	−0.30
matched:Q_high	−0.16	−2.20	0.093	n.s.	−0.22
matched:k_max	−0.14	−1.83	0.141	n.s.	−0.03
matched:alpha_max	−0.23	−2.15	0.098	n.s.	+0.07
matched:v_max	−0.29	−5.31	0.006	PINN better	−0.18
matched:corner	−0.33	−1.59	0.186	n.s.	−0.16

Table 15 The same strong-form PINN implemented twice, in PyTorch and in JAX, on the random and extrap:alpha_max splits, 5 seeds each: accuracy (macro-RMSE, MPa) and cost. Mean $\pm$ sd over seeds; p from a seed-paired t -test. Timings are for CPU, double precision.

Split	Backend	macro-RMSE, MPa	Wall time, s
random	PyTorch	7.98 ± 0.21	311 ± 118
	JAX	8.12 ± 0.18	557 ± 118
	paired p (JAX vs PyTorch)	$p = 0.269$	$p = 0.00036$
extrap:alpha_max	PyTorch	13.05 ± 0.49	316 ± 97
	JAX	13.49 ± 0.30	476 ± 110
	paired p (JAX vs PyTorch)	$p = 0.169$	$p = 0.047$
both splits	PyTorch	—	313 ± 102
	JAX	—	516 ± 150
	ratio JAX/PyTorch	—	$\times 1.65$
	paired p ($n = 10$)	—	$p = 0.002$

Table 16 Robustness to label corruption on the random split: macro-RMSE on the clean hold-out, MPa, mean $\pm$ sd over 3 seeds. Below the rule: the seed-paired gap to the data-driven baseline and its regression on the corrupted fraction.

Family	Corrupted training labels, %			Δ , MPa
	0	1	5	
MLP	8.19 ± 0.34	9.02 ± 0.34	12.00 ± 0.57	14.84 ± 0.15
PINN	8.09 ± 0.20	8.54 ± 0.09	10.89 ± 0.48	13.25 ± 0.69
VPINN	8.03 ± 0.19	9.13 ± 0.32	11.11 ± 0.04	14.34 ± 0.69
Δ (PINN – MLP), MPa	−0.10	−0.48	−1.11	−1.59
p (paired, 3 seeds)	0.718	0.100	0.010	0.037
Δ (VPINN – MLP), MPa	−0.16	+0.11	−0.89	−0.23
p (paired, 3 seeds)	0.478	0.747	0.123	0.253
Regression of the seed-level gap Δ (PINN – MLP) on the corruption fraction				

two-dimensional formulation with the axial coordinate as an input and the deliberate corruption of the training labels; 341 training runs were performed in total. On the results of the study the following conclusions can be drawn.

- (1) On the cleaned dataset of 1756 axisymmetric finite-element simulations, three surrogate families sharing everything but the loss reproduce the reference field to about 8 MPa of macro-RMSE on a random hold-out, with $R_n^2 = 0.99$ (global denominator) against 66 MPa for the constant predictor. The cost of prediction outside the training range is governed by the factor p with fixed n rather than by the family: measured against a random control of the same size, the error grows by a factor of 1.08 for the calibration ratio and 1.64 for the die semi-angle, but by 3.15 for the highest reduction and 5.93 for the highest drawing speed, the axis that coincides with the batch of simulations (Table 9).
- (2) A difference in accuracy between the three families was not established. The strong-form network has the lower macro-RMSE on all thirteen interpolation and extrapolation splits and the difference is significant on four of them in a seed-paired test, but it amounts to 0.3–1.9 MPa at an error level of 8–16 MPa, is of the order of the spread over seeds, and no difference from the data-driven baseline remains significant under a Holm correction over the thirteen splits (Table 14). A benefit of the physics prior specifically under a reduced training pool was not observed either: the slope of the family difference on the logarithm of the pool fraction is +0.20 MPa ($p = 0.21$) on the random split and −0.35 MPa ($p = 0.070$) on the semi-angle extrapolation, of opposite sign and significant on neither (Table 10). Both statements are therefore recorded as observations of a consistent direction and not as an established gain in accuracy.
- (3) The two software backends, PyTorch and JAX, are statistically indistinguishable in accuracy and differ only in cost, PyTorch completing a full training run 1.65 times faster on CPU in double precision (Table 15). The choice of backend is thus a matter of computational cost and not of the result obtained.
- (4) The contribution of the physics term lies in the admissibility of the predicted field and in the resistance to defects of the data source, not in accuracy. At equal accuracy the strong-form and full-form networks reduce the radial equilibrium residual and the violation of the traction-free condition well below those of the data-driven baseline, and, for the boundary condition, below the finite-element source itself (Table 11); the axial residual, by contrast, falls markedly only in the configuration whose loss contains the axial equation, so that this part of the gain is traceable to that term and not to the physics term in general (Table 12). As the training labels are corrupted, the gap in favour of the physics-informed model opens monotonically, by 14.2 MPa per unit corrupted fraction ($p = 0.00036$), whereas on clean labels the two are indistinguishable (Table 16).
- (5) The shear component τ_{rz} cannot be recovered from a one-dimensional, window-averaged representation: it changes sign along the axial coordinate, so that 79% of its magnitude is lost when the field is collapsed along z , which is a property of the representation and not of any model. A two-dimensional formulation with z as an input predicts this component with $R^2 = 0.93$ –0.94 (held-out denominator). The reference against which all of the above is measured is the finite-element solution itself. Its adequacy is addressed by the verification described in Section 3.1: on a subset of runs of the same model in which the exported frame is captured while the wire is still engaged in the die, the drawing stress obtained from the computed field is compared with 16 MPa parameters, with the 42 MPa analytical descriptions of the process – the slab-method relation of Siebel and the upper-bound solution of Avitzur. It is with respect

to a reference field verified in this way that the conclusions listed above are to be read. Further work is directed at extending the verification to the influence of the die semi-angle, for which the subset carrying the drawing load is too narrow, and at experimental validation of the residual-stress field, by X-ray diffraction for the surface stress and by layer removal for the through-thickness distribution.

Nomenclature

Q = reduction of diameter, $(D_0 - D)/D_0$
 k = calibration ratio (land length over outlet radius)
 L = length of the calibration land (m)
 r, z = radial and axial coordinates (m)
 R = billet radius (Fig. 4) or surface radius of the cross-section (Figs. 5–6)
 $\sigma_{rr}, \sigma_{\theta\theta}, \sigma_{zz}$ = normal residual stress components (MPa)
 τ_{rz} = shear residual stress component (MPa)
 R_r, R_z = radial and axial equilibrium residuals
 R_0, R_f = radius of the billet and of the drawn wire (m)
 v = drawing velocity (m/min)
 p = normal pressure on the die contact (MPa)
 m = friction factor, ratio of the interface shear stress to the shear yield stress
 $f(\alpha)$ = die-angle function of the Avitzur velocity field
 c = specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)
 T = temperature ($^{\circ}\text{C}$)

Greek Letters

α = half-angle of the die working cone (deg)
 μ = Coulomb friction coefficient
 σ_d = drawing stress (MPa)
 σ_f = flow stress (MPa)
 $\epsilon_{\text{hom}}, \epsilon_{\text{eff}}$ = homogeneous and effective strain
 $\dot{\epsilon}$ = mean strain rate over the pass (s^{-1})
 Δ = inhomogeneity parameter of the deformation zone
 Φ = excess-work factor
 β = fraction of plastic work converted to heat
 ρ = density (kg m^{-3})

Abbreviations

MLP = multilayer perceptron (data-driven surrogate)
PINN = physics-informed neural network (strong form)
VPIINN = variational physics-informed neural network (weak form)
FEM = finite-element method

References

- [1] Burioli, S. et al., 2024, "A novel numerical approach for analyzing experimental data on critical current degradation in Nb_3Sn wires caused by transverse deformations preceding heat treatment," *Superconductor Science and Technology*, **37**(9), p. 095007.
- [2] Lim, J., Lee, J., and Jang, B., 2026, "A Study of Effect of Bidirectional Drawing on the Mechanical Properties of 30MnSi6 Non-Heat-Treated Steel," *Metals*, **16**(1), p. 118.
- [3] Shin, Y. et al., 2026, "Accurate Finite Element Simulation of Unidirectional and Alternate Multi-Pass Drawings, Focusing on Residual Stresses," *Applied Sciences*, **16**(4), p. 2154.
- [4] del Rey, J. C. et al., 2025, "Effect of Die Design and Lubricant on ZnAl15% Wire Drawing: An Experimental Approach with Pressure/Hydrodynamic and Conventional Drawing Dies," *Lubricants*, **13**(11), p. 481.
- [5] Del Rey, J. C. et al., 2026, "Advances on the optimization of the thermal treatment and modelization on the multi-stage wire drawing process with ZnAl15% alloy wires," *CIRP Journal of Manufacturing Science and Technology*, **67**, pp. 29–47.
- [6] He, S., Van Bael, A., Li, S., Van Houtte, P., Mei, F., and Sarban, A., 2003, "Residual stress determination in cold drawn steel wire by FEM simulation and X-ray diffraction," *Materials Science and Engineering: A*, **346**(1–2), pp. 101–107.
- [7] Överstam, H., 2006, "The influence of bearing geometry on the residual stress state in cold drawn wire, analysed by the FEM," *Journal of Materials Processing Technology*, **171**(3), pp. 446–450.
- [8] Aghdami, A. M. and Davoodi, B., 2020, "An inverse analysis to identify the Johnson-Cook constitutive model parameters for cold wire drawing process," *Mechanics & Industry*, **21**(5), p. 527.
- [9] Radionova, L., Lisovskiy, R., Svistun, A., Gromov, D., and Erdakov, I., 2022, "FEM simulation analysis of wire drawing process at different angles dies on straight-line drawing machines," *International Conference on Industrial Engineering*, Springer, pp. 769–778.
- [10] Selbmann, R. et al., 2024, "Residual Stress Engineering for Wire Drawing of Austenitic Stainless Steel X5CrNi18-10 by Variation in Die Geometries—Effect of Drawing Speed and Process Temperature," *Materials*, **17**(5), p. 1174.
- [11] Suliga, M. et al., 2024, "The Influence of Temperature in the Wire Drawing Process on the Wear of Drawing Dies," *Materials*, **17**(20), p. 4949.
- [12] Rumiński, M., Skubisz, P., and Micek, P., 2025, "Effect of reduction ratio on die fill-out and hardness profile of cold-drawn polygonal rods made of acid resistant steel X6CrNiTi18-10," *Journal of Mining and Metallurgy, Section B: Metallurgy*, **61**(2), pp. 235–250.
- [13] Mozaffar, M. et al., 2022, "Mechanistic artificial intelligence (mechanistic-AI) for modeling, design, and control of advanced manufacturing processes: Current state and perspectives," *Journal of Materials Processing Technology*, **302**, p. 117485.
- [14] Goodfellow, I., Bengio, Y., and Courville, A., 2016, *Deep Learning*, MIT Press, Cambridge, MA.
- [15] Weichert, D., Link, P., Stoll, A., Rüping, S., Ihlenfeldt, S., and Wrobel, S., 2019, "A review of machine learning for the optimization of production processes," *The International Journal of Advanced Manufacturing Technology*, **104**(5), pp. 1889–1902.
- [16] Gromov, D. V., Radionova, L. V., and Lisovskaya, T. A., 2026, "Residual Stress Prediction in Wire Drawing by using Integrated Artificial Neural Networks and Finite Element Analysis," *2026 International Russian Smart Industry Conference (SmartIndustryCon)*, IEEE, pp. 1147–1152.
- [17] Demin, D., Grebenkin, I., and Barinov, A., 2025, "Forecasting the residual stress components in wires using an artificial neural network," *The International Journal of Advanced Manufacturing Technology*, pp. 1–12.
- [18] Demin, D. and Grebenkin, I., 2024, "Application of ML methods to predict residual stresses and strains after wire drawing process," *International Journal of Advanced Manufacturing Technology*, **133**(7), pp. 3461–3473.
- [19] Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M., 2019, "Optuna: A Next-generation Hyperparameter Optimization Framework," *1907.10902*, <https://arxiv.org/abs/1907.10902>
- [20] Park, M. et al., 2026, "Indentation-informed convolutional neural network for simultaneous prediction of non-equibiaxial residual stress and plastic flow," *International Journal of Plasticity*, p. 104606.
- [21] Han, G. et al., 2024, "Evaluation of plastic properties and equi-biaxial residual stress via indentation and ANN," *Materials & Design*, **239**, p. 112745.
- [22] Wang, Y. et al., 2025, "An online prediction method of three-dimensional machining residual stress field based on inceptu-net," *Measurement*, **242**, p. 115794.
- [23] Tsui, H. S. R., 2026, "Metrology-guided estimation and inverse design of drawing force in cold rod drawing using a FEM-ANN-XGBoost hybrid," *Measurement*, **269**, p. 120783.
- [24] Raissi, M., Perdikaris, P., and Karniadakis, G. E., 2019, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, **378**, pp. 686–707.
- [25] Karniadakis, G. E., Kevrekidis, I. G., Lu, L., Perdikaris, P., Wang, S., and Yang, L., 2021, "Physics-informed machine learning," *Nature Reviews Physics*, **3**, pp. 422–440.
- [26] Haghighat, E., Raissi, M., Moure, A., Gomez, H., and Juanes, R., 2021, "A physics-informed deep learning framework for inversion and surrogate modeling in solid mechanics," *Computer Methods in Applied Mechanics and Engineering*, **379**, p. 113741.
- [27] Ra, D. et al., 2026, "Physics-informed machine learning across manufacturing processes: Recent advances, challenges, and directions," *Journal of Manufacturing Systems*, **85**, pp. 72–95.
- [28] Yang, T. et al., 2025, "Prediction of residual stress field of ultrasonic rolling processed 20CrNiMo based on physics-informed neural networks," *Journal of Manufacturing Processes*, **149**, pp. 841–851.
- [29] Mehedi Hasan, M. and Schoop, J., 2026, "Inverse Parameter Identification of Subsurface Residual Stress in Tractional Sliding Processes Using a Physics-Informed Neural Network," *Journal of Tribology*, **148**(6), p. 061501.
- [30] Hasan, M. M., 2025, "Machining-induced Residual Stress Modeling Using Physics-Informed Neural Networks," .
- [31] Muransky, O., Tran, M. N., and Payten, W., 2026, "On the application of PINN-style physics-regularised neural networks to high-temperature creep rupture life prediction," *International Journal of Pressure Vessels and Piping*, **219**.
- [32] Munzone, F. et al., 2024, "Comparative study of artificial neural network and physics-informed neural network application in sheet metal forming," *27th International ESAFORM Conference on Material Forming (ESAFORM 2024)*, Association of American Publishers, pp. 2278–2288.
- [33] Laakso, S. V. A. et al., 2024, "Hybrid FE-ML model for turning of 42CrMo4 steel," *CIRP Journal of Manufacturing Science and Technology*, **55**, pp. 333–346.
- [34] Rahman, A. et al., 2025, "Physics-based machine learning framework for predicting structure-property relationships in DED-fabricated low-alloy steels," *Metals*, **15**(9), p. 965.
- [35] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai,

- J., and Chintala, S., 2019, "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *Advances in Neural Information Processing Systems* 32, pp. 8024–8035.
- [36] Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., and Zhang, Q., 2018, "JAX: Composable Transformations of Python+NumPy Programs," <http://github.com/google/jax>.
- [37] Lu, L., Meng, X., Mao, Z., and Karniadakis, G. E., 2021, "DeepXDE: A deep learning library for solving differential equations," *SIAM Review*, **63**(1), pp. 208–228.
- [38] Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., and Dormann, C. F., 2017, "Cross-Validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure," *Ecography*, **40**(8), pp. 913–929.
- [39] Dassault Systèmes Simulia, 2022, *Abaqus Analysis User's Guide*, Dassault Systèmes, Providence, RI, USA.
- [40] Tancik, M., Srinivasan, P. P., Mildenhall, B., Fridovich-Keil, S., Raghavan, N., Singhal, U., Ramamoorthi, R., Barron, J. T., and Ng, R., 2020, "Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains," *Advances in Neural Information Processing Systems* 33, pp. 7537–7547.
- [41] Wang, S., Yu, X., and Perdikaris, P., 2022, "When and why PINNs fail to train: A neural tangent kernel perspective," *Journal of Computational Physics*, **449**, p. 110768.
- [42] Kharazmi, E., Zhang, Z., and Karniadakis, G. E., 2021, "hp-VPINNs: Variational physics-informed neural networks with domain decomposition," *Computer Methods in Applied Mechanics and Engineering*, **374**, p. 113547.
- [43] Loshchilov, I. and Hutter, F., 2019, "Decoupled Weight Decay Regularization," *International Conference on Learning Representations (ICLR)*.
- [44] Demšar, J., 2006, "Statistical comparisons of classifiers over multiple data sets," *Journal of Machine Learning Research*, **7**, pp. 1–30.
- [45] Siebel, E., 1947, "Der Derzeitige Stand der Erkenntnisse über die Mechanischen Vorgänge beim Drahtziehen," *Stahl und Eisen*, **66/67**(11/12), pp. 171–180.
- [46] Avitzur, B., 1968, *Metal Forming: Processes and Analysis*, McGraw-Hill, New York, Reprinted by Robert E. Krieger, Huntington, NY, 1979.
- [47] Avitzur, B., 1971, "Study of Flow Through Conical Converging Dies," *Metal Forming: Interrelation Between Theory and Practice*, A. L. Hoffmann, ed., Plenum Press, New York, pp. 1–46.
- [48] Vega, G., Haddi, A., and Imad, A., 2009, "Temperature Effects on Wire-Drawing Process: Experimental Investigation," *International Journal of Material Forming*, **2**(S1), pp. 229–232.
- [49] Vega, G., Haddi, A., and Imad, A., 2009, "Investigation of Process Parameters Effect on the Copper-Wire Drawing," *Materials & Design*, **30**(8), pp. 3308–3312.
- [50] Haddi, A., Imad, A., and Vega, G., 2011, "Analysis of Temperature and Speed Effects on the Drawing Stress for Improving the Wire Drawing Process," *Materials & Design*, **32**(8–9), pp. 4310–4315.
- [51] Backofen, W. A., 1972, *Deformation Processing*, Addison-Wesley, Reading, MA, USA.
- [52] Holm, S., 1979, "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics*, **6**(2), pp. 65–70.
- [53] Friedman, M., 1937, "The use of ranks to avoid the assumption of normality implicit in the analysis of variance," *Journal of the American Statistical Association*, **32**(200), pp. 675–701.

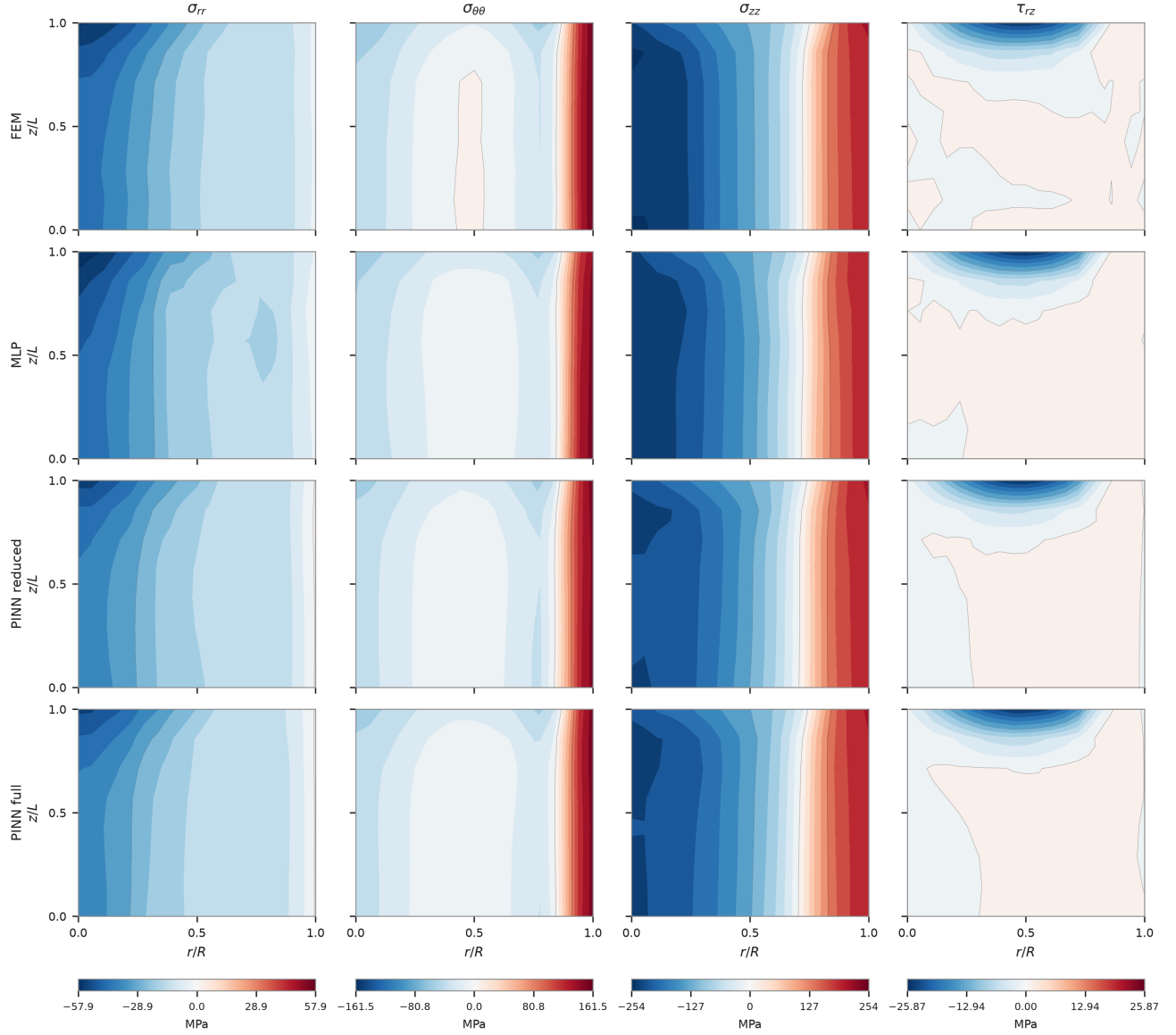


Fig. 5 Distribution of the residual stress tensor components over the investigated region for FEM, MLP, and PINN in the full and reduced formulations at $Q = 0.025$ (2.5% in diameter, 5.4% in area), $k = 0.1$, $\alpha = 8^\circ$, $\mu = 0.10$, $v = 20$ m/min. Rows: FEM reference and the three model families; columns: σ_{rr} , $\sigma_{\theta\theta}$, σ_{zz} , τ_{rz} . The colour scale is common to all rows within a component. r is normalised by the surface radius in the same cross-section ($r/R = 1$ is the free surface), z by the window length. The case shown has the median test error of the full PINN.

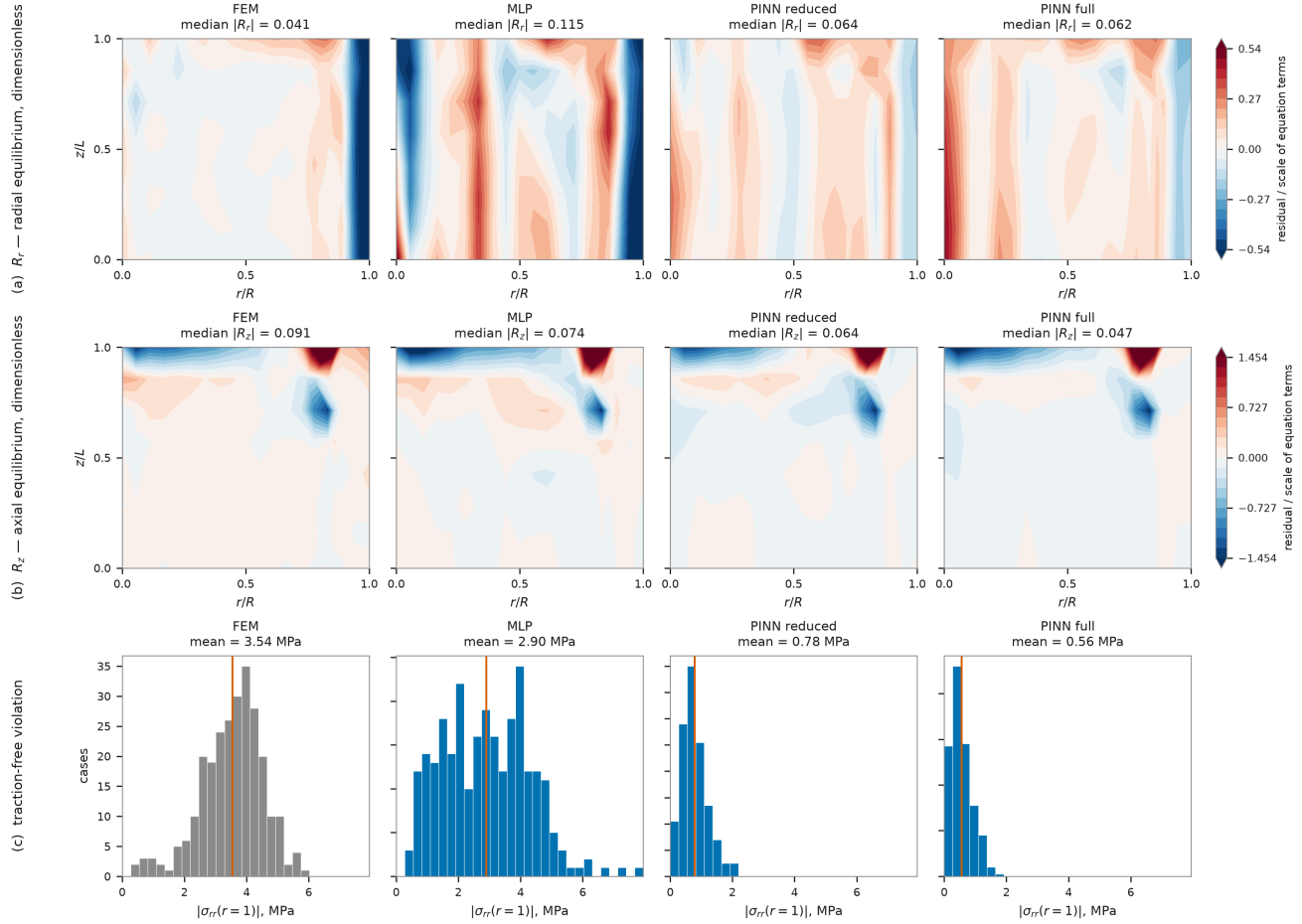


Fig. 6 Comparison of the equilibrium residual distributions R_r and R_z over the investigated region for FEM, MLP, and PINN in the full and reduced formulations at $Q = 0.025$, $k = 0.1$, $\alpha = 8^\circ$, $\mu = 0.10$, $v = 20$ m/min (same case as Fig. 5). (a) radial residual R_r , (b) axial residual R_z , both made dimensionless by the characteristic magnitude of the terms entering the respective equation; the panel titles give the median $|R|$ over all 261 test cases. (c) distribution over the test cases of the traction-free violation $|\sigma_{rr}|$ at $r/R = 1$, mean marked. The FEM column is the reference: the source itself does not satisfy the equations exactly.

List of Figures

1	Axisymmetric wire-drawing process. S_1 – axis of symmetry; S_2 – outlet face of the finished product; S_3 – free lateral surface; S_4 – contact surface with the die; S_5 – entry face of the billet; α – half-angle of the working cone of the die; D_0 and D – diameters of the billet and of the drawn wire; L – length of the calibration land	3
2	Training of the data-driven MLP surrogate. The architecture and every hyperparameter are fixed before any held-out region is evaluated; the loop shown is the training loop of a single run, and no hyperparameter search surrounds it.	5
3	Training of the physics-informed surrogate: shared trunk, equilibrium residual and traction-free boundary term in the loss. The loss weights are fixed a priori; the only quantity read from the data is the gradient-norm calibration g of Eq. (8), set once on the first mini-batch.	6
4	Distribution of residual stresses obtained from the FEM simulation at $Q = 0.10$ (10 % reduction in diameter, 20.1 % in area), $k = 0.5$, $\alpha = 12^\circ$, $\mu = 0.05$, $v = 10$ m/min. Panels from left to right: the components σ_{rr} , $\sigma_{\theta\theta}$, σ_{zz} and τ_{rz} over the full axial extent of the wire, each with its own colour scale graduated in MPa. The radius is normalised by the billet radius $R = 18$ mm and the axial coordinate by the model length $L = 139.5$ mm; dashed lines mark the axial window 25–75 % of L from which the training data are taken.	8
5	Distribution of the residual stress tensor components over the investigated region for FEM, MLP, and PINN in the full and reduced formulations at $Q = 0.025$ (2.5 % in diameter, 5.4 % in area), $k = 0.1$, $\alpha = 8^\circ$, $\mu = 0.10$, $v = 20$ m/min. Rows: FEM reference and the three model families; columns: σ_{rr} , $\sigma_{\theta\theta}$, σ_{zz} , τ_{rz} . The colour scale is common to all rows within a component. r is normalised by the surface radius in the same cross-section ($r/R = 1$ is the free surface), z by the window length. The case shown has the median test error of the full PINN.	17
6	Comparison of the equilibrium residual distributions R_r and R_z over the investigated region for FEM, MLP, and PINN in the full and reduced formulations at $Q = 0.025$, $k = 0.1$, $\alpha = 8^\circ$, $\mu = 0.10$, $v = 20$ m/min (same case as Fig. 5). (a) radial residual R_r , (b) axial residual R_z , both made dimensionless by the characteristic magnitude of the terms entering the respective equation; the panel titles give the median $ R $ over all 261 test cases. (c) distribution over the test cases of the traction-free violation $ \sigma_{rr} $ at $r/R = 1$, mean marked. The FEM column is the reference: the source itself does not satisfy the equations exactly.	18

List of Tables

1	Ranges of process parameters in the FEM campaign.	3
2	Share of the variance along z inside the axial window and effect of collapsing the field along z by the median (mean absolute value, MPa).	5
3	Case-level filtering applied before splitting.	5
4	The 13 case-level splits of the 1756 cases. n_{train} is the training pool, including the early-stopping subset.	6
5	Model families. Within each track all families share the network, the optimiser, the training conditions (epoch limit, early-stopping rule, mini-batch size), the seeds and the splits; only the loss composition differs. R_1 : reduced radial residual (3); R_r, R_z : full residuals (1)–(2); W_k : weak residual (9).	8
6	Drawing stress (MPa) for the four runs: the finite-element value against the Siebel relation, Eq. (10), the Avitzur upper bound in its friction-factor form, Eq. (11) with $m = \sqrt{3}\mu$, and the modified Avitzur relation in the Coulomb form of Eq. (12). All three use the flow stress derived from the constitutive equation, 300–305 MPa, and no fitted parameter. Reduction Q is nominal, in diameter; the achieved outlet radius is 17.71 mm in every run. The last two rows are the mean absolute deviation from the finite-element value and the secant slope $d\sigma_d/d\mu$ between the $\mu = 0.025$ and $\mu = 0.1$ rows at $\alpha = 12^\circ$, for which the finite-element value is 323 MPa.	10
7	Intermediate quantities of the closed self-consistent estimate, Eqs. (14)–(15), for the four runs: inhomogeneity parameter Δ , Backofen excess-work factor Φ , effective strain ε_{eff} , mean strain rate $\dot{\varepsilon}$, flow stress σ_f from the constitutive equation at the fixed-point temperature, and the deformation part of the drawing stress, σ_d^{def} , being the ideal work, the redundant shear and the friction on the cone, with the friction on the land excluded for the reason given in the text. Nothing is taken from the finite-element model. The nominal geometry is used throughout, $\varepsilon_{\text{hom}} = 0.032$ at $R_f = 17.71$ mm, so Δ here is the value for that geometry; evaluated instead at the outlet radius measured in each run it is 25.2–25.7 and 17.0.	10
8	Mean temperature rise during the pass and the resulting pass-mean temperature: the self-consistent analytical estimate, in which the deformation work and the temperature close on each other and nothing is read from the model, against the cross-section average of the exported temperature history. Every run starts at 20 °C, and the pass-mean temperature is formed from the measured rise. The peak rise, reached at the surface, is 45.7–74.8 °C, three to four times the section mean. Reduction 1.6 % in diameter (3.2 % in area), land ratio unity.	10
9	Cost of holding out a whole factor level, averaged over the three families: macro-RMSE on the held-out region, on the matched control of the same size, their ratio, R_n^2 (global denominator) on the held-out region, and the constant predictor (MPa).	12
10	Learning curves: macro-RMSE (MPa) versus the fraction of the training pool, three seeds. Δ is PINN minus MLP with the seed-paired p -value. Slope of Δ on $\log(\text{fraction})$: +0.20 MPa ($p = 0.21$) on random, –0.35 MPa ($p = 0.070$) on <code>extrap:alpha_max</code> .	12
11	Accuracy and physical admissibility of the three two-dimensional families on a common hold-out of 261 cases. Accuracy: component RMSE (MPa)/ R^2 and macro-RMSE; admissibility: the dimensionless equilibrium residuals $ R_r $, $ R_z $ and the traction-free violation $ \sigma_{rr}(r=1) $, MPa. Definitions, denominators and the training pool are given in the text.	13
12	Ablation on the extended track (z as input, 8×20 grid), three seeds: macro-RMSE (MPa) and median full-form equilibrium residuals (MPa/m) on the held-out cases; in brackets, change relative to the MLP with the same input. The three configurations differ only in the physics terms of the loss.	13
13	Macro-averaged RMSE on the held-out cases for the 13 splits of the main track, MPa, mean $\pm$ sd over 5 seeds. Split types are defined in Section 2.2.	13
14	Seed-paired comparison of the physics-informed families with the data-driven baseline on every split of Table 13. Δ is the mean over seeds of the per-seed difference in macro-RMSE, MPa; negative favours the physics-informed model. Holm correction over the 13 splits, p -value in brackets. $p < 0.05$ are significant.	14
15	The same strong-form PINN implemented twice, in PyTorch and in JAX, on the random and <code>extrap:alpha_max</code> splits, 5 seeds each: accuracy (macro-RMSE, MPa) and cost. Mean $\pm$ sd over seeds; p from a seed-paired t -test. Timings are for CPU, double precision.	14
16	Robustness to label corruption on the random split: macro-RMSE on the clean held-out, MPa, mean $\pm$ sd over 3 seeds.	14